Controlled Inversion of Image-Based Time-Series Encodings: Operational Decoders for GAF, RP, and MTF Representations

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Image-based encodings of time series — Recurrence Plots (RP), Markov Transition Fields (MTF), and the summation (GASF) and difference (GADF) variants of Gramian Angular Fields — expose temporal structure to image-domain models [9, 10]. This work does not evaluate generative augmentation; it studies the prerequisite inverse problem: whether images from known encoders can be mapped back to triaxial accelerometer windows with controlled loss. We contribute three operational decoders with, to our knowledge, no prior fully specified counterpart. The first is a signed-RP decoder that, decoding each image on its own continuous RP distance scale, recovers the two-dimensional adjacent-state geometry through classical multidimensional scaling, enforces the consecutive-state overlap, and uses the Lu–Tong sign convention to select the remaining orientation — resolving, for the *continuous signed* RP of overlapping states, the reflection ambiguity that the graph-geodesic reconstruction of Hirata, Horai and Aihara [7] leaves in its multidimensional-scaling embedding. The second is a closed-form, multi-pivot, boundary-anchored GADF decoder with a fixed arcsin branch, image-derived anchor and sign convention. The third is a Fiedler-vector spectral MTF decoder. Alongside these, we retain a GASF diagonal inverse [11] as a controlled prior-art baseline.

Under a unified four-lens protocol (visual, pointwise error, polynomial trend, and six-classifier original-trained robustness) on WISDM windows ($N = 128$), evaluated under both window-level and grouped subject-out 3-fold splits, RP, GASF and GADF form a single near-original reconstruction tier. Their global pointwise errors over all 8172 windows (the three window-level test folds pooled — the full dataset, since the decoders are deterministic and untrained) are MAE = 0.0148, 0.0205 and 0.0157 respectively (Pearson ≈ 1.0 in all three), and under the pooled paired McNemar analysis none shows a detected classification difference from the raw signal or from one another ($p_{\text{Holm}} = 1.0$ across all six classifiers); a subject-clustered TOST ($N = 51$ subjects) further confirms their equivalence to the original within a two-percentage-point accuracy margin, while MTF is equivalent to none. The three reach this tier through different inverses — RP through geometric multidimensional-scaling reconstruction of the adjacent states, GASF through its main diagonal, and GADF through off-diagonal angular differences — so we report empirical near-equivalence for this task rather than formal information preservation. MTF is the only many-to-one decoder: orders of magnitude behind on pointwise error and significantly outranked on every classifier, it is valuable as a compact classifier-oriented representation rather than a faithful inverse. Because every decoder assumes known encoder settings, this is controlled reconstruction — a necessary step toward, not a proof of, generative augmentation.

Keywords: Markov Transition Fields, Recurrence Plots, Gramian Angular Fields, Gramian Angular Difference Field, Time Series Reconstruction, Image Encoding, Accelerometer Data, Machine Learning.

1. Introduction

Time series data play a crucial role in fields ranging from finance and healthcare to human activity recognition. Despite their importance, many applications are constrained by scarce labelled data, which has motivated interest

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in data augmentation and, more broadly, in alternative representations that can leverage advanced methods from adjacent domains — most notably the rich domain of image processing.

A common approach is to convert time-series signals into image-based representations. Techniques such as Markov Transition Fields (MTF), Recurrence Plots (RP), and Gramian Angular Fields (GAF) transform a one-dimensional sequence into a two-dimensional matrix whose pixel (i, j) encodes a relation between two time indices. For triaxial accelerometer windows, the three axes are encoded independently and then treated as image channels. This construction allows image analysis and generation models to operate on temporal structure that is often difficult to expose directly in the raw signal.

Parallel to these developments, diffusion models have gained prominence in image generation. Their ability to synthesize highly structured images motivates the use of image representations for time-series augmentation, but this paper does not train or evaluate a diffusion or GAN model. We focus on the necessary inverse problem that must be solved first: if an image is produced by a known time-series encoder, can it be reconstructed back into the signal domain with controlled loss?

Our approach addresses this inverse problem by proposing dedicated reconstruction strategies for RP, MTF, and GAF-based encodings, and by evaluating the completed decoders under a unified framework that includes visual inspection, reconstruction metrics, polynomial trend comparison, and downstream classification. The overall workflow is summarized in Fig. 1. Experiments on the WISDM Smartphone and Smartwatch Activity and Biometrics Dataset [14] show that the evaluated encodings preserve different aspects of the original signals and therefore support different use cases.

A central narrative choice in this paper is to separate *controlled reconstruction* from *unconstrained generative augmentation*. The experiments decode images produced by known encoders with known encoder settings and, when physical units are reported, use per-window min–max calibration recorded during encoding. This setting is the necessary first test of whether an image representation contains enough information to return to the time domain. It is not yet a claim that arbitrary images produced by a generative model will be physically plausible or reconstructable without additional calibration.

The paper is organized around three questions: what information each image encoding preserves, how that information can be mapped back to the time domain, and whether the reconstructed signals remain useful for downstream activity recognition. The main contributions are:

- a common operator-level formulation $M_{ij} = \omega(\xi(x_i), \xi(x_j))$ that, in the discussion, lets us read signed RP and GADF as structurally adjacent parametrisations of directional pairwise differences — GADF lying in the antisymmetric zero-diagonal matrix space by construction, signed RP approximating it without exact antisymmetry under the Lu–Tong sign rule;
- **a novel RP decoder** that recovers the full two-dimensional adjacent-state geometry from the *continuous signed* Lu–Tong RP of overlapping states — the specific reconstruction problem not addressed by Hirata–Horai–Aihara [7], whose graph-geodesic multidimensional scaling reconstructs the original series from unsigned recurrence plots but leaves its reflection ambiguity unresolved: decoding the image on the RP matrix's own distance range (a round-trip correctness precondition, not a competing method), it recovers the adjacent states by classical multidimensional scaling, enforces the consecutive-state overlap, and uses the signed-RP convention to select the remaining orientation — the ablation of Section 6.6 shows that, once the round-trip is correct, the two algorithmic choices, retaining the full two-dimensional state geometry and resolving its orientation through the signed RP, are what make RP reconstruction near-lossless (MAE ≈ 0.015, Pearson ≈ 1.0);
- **a novel restricted-branch, multi-pivot, boundary-anchored GADF decoder** that operationalises the earlier observation by Wang and Oates [9] that GADF can admit an accurate inverse map under restricted scaling: the contribution is a fully specified closed-form decoder with an explicit arcsin branch condition, image-derived angular anchor, fixed sign convention, optional min–max calibration for physical-scale recovery, and empirical evaluation under the same protocol as RP, MTF, and GASF, so it should not be read as a generic inverse for unconstrained GADF images;
- **a novel Fiedler-vector MTF decoder**, for which we are not aware of a prior explicit MTF-to-time-series decoder: rows of the field are clustered into Q bins, the bin transition matrix is recovered, and the bin ordering is fixed by the Fiedler vector of the corresponding graph Laplacian; because MTF is quantised and many-to-one, the output is a representative compatible quantile trajectory rather than a bijective inverse;
- a GASF inversion pipeline that extends the diagonal-inverse line of work [11] with explicit handling of normalisation and quantisation, kept in the comparison as a controlled prior-art baseline rather than as a novel decoder;
- an explicit account of the assumptions that make each inverse map exact, approximate, or heuristic, and the encoder settings or optional calibration that each one requires;

- an evaluation design that separates pointwise fidelity, trend-level similarity, and classification utility, so that good classification is not mistaken for accurate signal reconstruction;
- a fully comparable evaluation of all four encodings (RP, MTF, GASF, GADF) on the same evaluation set — the pointwise and polynomial-trend metrics computed over all 8172 windows (the three test folds pooled, i.e. the full dataset scored once, since the decoders are deterministic and untrained), and the same six-classifier panel evaluated on those same pooled windows via the per-fold models — reported under both the stratified window-level and grouped subject-out 3-fold protocols.

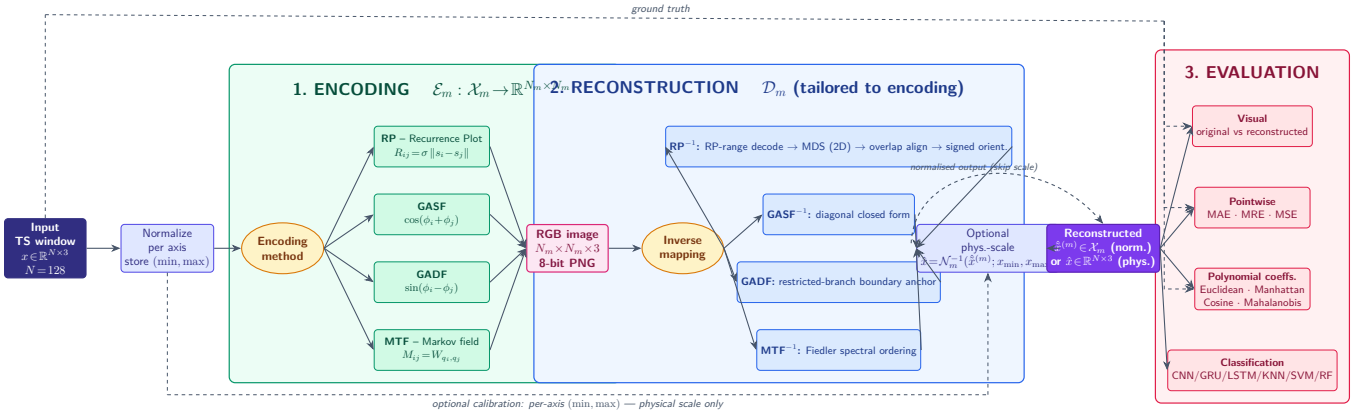


Figure 1: Controlled image-based time-series reconstruction workflow. Solid arrows show the signal/image path from the triaxial window $x \in \mathbb{R}^{N\times 3}$ through per-axis encoding ($\mathcal{E}_m : \mathcal{X}_m \to \mathbb{R}^{N_m \times N_m}$, with $N_m = N$ for GASF/GADF/MTF and $N_m = N-1$ for RP) and method-tailored inversion ($\mathcal{D}_m$), which returns a normalised reconstruction $\hat{\tilde{x}}^{(m)} \in \mathcal{X}_m$. The final block is dual: the decoder output can be consumed directly in the normalised domain, or passed through the *optional* physical-scale recovery $\mathcal{N}_m^{-1}$ to obtain $\hat{x}$ in accelerometer units. Dashed arrows show auxiliary information recorded during encoding: the per-axis $(\min, \max)$ calibration, used *only* when physical-scale recovery is requested, and the original window used as ground truth for visual, pointwise, and trend metrics. The RP decoder additionally consumes its own per-axis RP-matrix range $(R_{\min}, R_{\max})$ — needed even for the normalised reconstruction and distinct from the physical-scale $(\min, \max)$ — omitted from the arrows to keep the diagram legible.

2. Related Work and Encodings

This section fixes the notation and summarizes the encoding mechanisms needed for the inverse problem. We distinguish between the image construction rule, which is part of the prior literature, and the reconstruction rule, which is where this paper contributes or identifies an explicit gap. For clarity, GAF denotes the family, whereas GASF and GADF denote the summation and difference variants, respectively.

All four encodings can be written as per-axis operators $\mathcal{E}_m : \mathcal{X}_m \to \mathbb{R}^{N_m \times N_m}$, but they preserve different kinds of information. RP encodes geometric recurrences in a phase-space embedding, GASF and GADF encode angular relations after polar normalization, and MTF encodes transition probabilities between quantised states. This distinction is important for the rest of the paper: a representation that is useful for classification is not necessarily pointwise invertible, and a representation that is highly invertible under controlled conditions may still require metadata that are difficult to obtain for generated images.

Table 1: Operator-level view of the four image encodings.

Method	Operator	Main Information Preserved	Inverse Character
RP	$\mathrm{sign}(i,j)\,\lVert s_i - s_j \rVert$	Phase-space geometry and recurrence	Near-lossless: 2-D MDS of the RP-range distances, overlap alignment, signed orientation
GASF	$\cos(\phi_i + \phi_j)$	Absolute angular magnitude via the diagonal	Closed form under a fixed branch
GADF	$\sin(\phi_i - \phi_j)$	Relative angular differences	Closed form with restricted scaling and anchor
MTF	W_{q_i,q_j}	Markov transition statistics between bins	Heuristic, quantised reconstruction

Table 2 positions each decoder against the closest prior reconstruction work and states the specific gap it closes, so that the novelty claims made throughout the paper can be checked against a single reference point. For GADF and MTF we are not aware of any prior fully specified time-series inverse; for RP, Hirata–Horai–Aihara [7] reconstruct the original series from unsigned recurrence plots, but no prior work we reviewed reconstructs the scalar sequence from the *continuous signed* RP of overlapping states, which is the specific inverse this paper specifies. GASF is the exception and is kept only as a controlled prior-art baseline.

Table 2: Closest prior reconstruction work per encoding and the gap this paper closes. "Prior inverse to the time series?" asks whether earlier work recovered the signal itself, not merely restored or classified the image.

Method	**Closest prior reconstruction work**	**Prior inverse to the time series?**	**What this paper adds**
RP	Hirata–Horai–Aihara [7]; Thiel [5]; Tanio et al. [6]	Yes, but from *unsigned* RPs via graph-geodesic MDS, recovered only up to a reflection	For the *continuous signed* RP of overlapping states: two-dimensional MDS state recovery, consecutive-state overlap alignment, and signed-RP orientation selection that resolves the reflection, decoded on the RP's own distance range — near-lossless (MAE ≈ 0.015, Pearson ≈ 1.0)
GASF	Huamin et al. [11] (diagonal half-angle)	Yes, from the main diagonal	Re-implemented as a controlled prior-art baseline with explicit normalisation/quantisation handling (not claimed as novel)
GADF	Wang and Oates [9] note invertibility under $[0, 1]$ scaling	Stated but not specified: their practical route goes through the GASF diagonal	Explicit closed-form multi-pivot, boundary-anchored decoder with fixed arcsin branch, image-derived anchor and sign convention (min–max only for optional physical-scale recovery)
MTF	No prior MTF→series decoder; MRF work [12, 13] restores images, not series	No	Fiedler-vector spectral bin-ordering yielding a representative compatible quantile trajectory

2.1. Recurrence Plot

2.1.1. Image Construction

Recurrence plots were introduced by Eckmann et al. [1] and are commonly interpreted through phase-space reconstruction ideas developed for nonlinear dynamical systems [2, 3]. A scalar time series can be embedded as delay vectors

$$\mathbf{s}_t = (x_t, x_{t+\tau}, \ldots, x_{t+(m-1)\tau}) \in \mathbb{R}^m,$$

where m is the embedding dimension and τ is the delay. In this work, we use the first-order embedding $m = 2$, $\tau = 1$, so $s_i = (x_i, x_{i+1})$. The classical recurrence matrix is then defined as

$$\mathbf{R}_{i,j}(\epsilon) = \boldsymbol{\theta}(\epsilon - \|s_i - s_j\|), \quad i, j = 1, \ldots, N-1, \tag{1}$$

where ϵ is a threshold that determines the neighborhood size, $\boldsymbol{\theta}$ is the Heaviside step function, and $\|\cdot\|$ denotes a norm. Choosing an appropriate norm is crucial for constructing a meaningful recurrence plot; the most commonly used norms are L_1, L_2, and L_∞ [4].

One limitation of this definition is that the distance from x_i to x_j equals that from x_j to x_i, yielding a symmetric matrix with respect to the main diagonal. Consequently, the mapping is not bijective: a time series and its inverted version produce identical images, meaning that decoding the image would yield two possible time-series representations. To overcome this issue, we adopt the alternative definition proposed by Jianjie Lu and Kai-Yu Tong [4], which addresses the bijective image representation problem and is formulated as follows:

Consider a time series $x \in \mathbb{R}^{1\times N}$, where x_j denotes the j-th element of x. The states are then defined as

$$s_j = \big(x_j,\, x_{j+1}\big), \quad \text{with } s_j \in \mathbb{R}^2,$$

so that the recurrence plot (RP) is constructed as a matrix $R \in \mathbb{R}^{(N-1)\times(N-1)}$.

A basic formulation of the RP is given by

$$R_{m,n} = \|s_m - s_n\|, \quad m, n = 1, \ldots, N-1, \tag{2}$$

where $\|\cdot\|$ denotes a chosen norm (e.g., L_1, L_2, or L_∞).

To incorporate directional information, we calculate a sign based on the differences between state vectors. Throughout this work we use the fixed reference direction $v = (1, 1)^\top / \sqrt{2}$. For off-diagonal entries, define

$$\gamma_{mn} = \frac{(s_m - s_n) \cdot v}{\|s_m - s_n\| \, \|v\|}, \qquad m \neq n.$$

The sign convention assigns a negative value only when the off-diagonal displacement vector forms an angle greater than $3\pi/4$ with respect to v:

$$\operatorname{sign}(m, n) = \begin{cases} 1, & \text{if } m = n, \\ -1, & \text{if } m \neq n \text{ and } \gamma_{mn} < \cos\left(\dfrac{3\pi}{4}\right), \\ 1, & \text{otherwise,} \end{cases} \tag{3}$$

where $\|v\| = 1$. The diagonal convention avoids the undefined ratio at $m = n$; since $R_{m,m} = 0$ after multiplication by the distance, the chosen diagonal sign has no numerical effect. This convention is used consistently during both RP encoding and the reconstruction-orientation check.

Finally, the recurrence plot is redefined as

$$R_{m,n} = \operatorname{sign}(m, n) \cdot \|s_m - s_n\|, \tag{4}$$

which is the definition we use to encode the RP as an image after normalizing the values and mapping them to RGB.

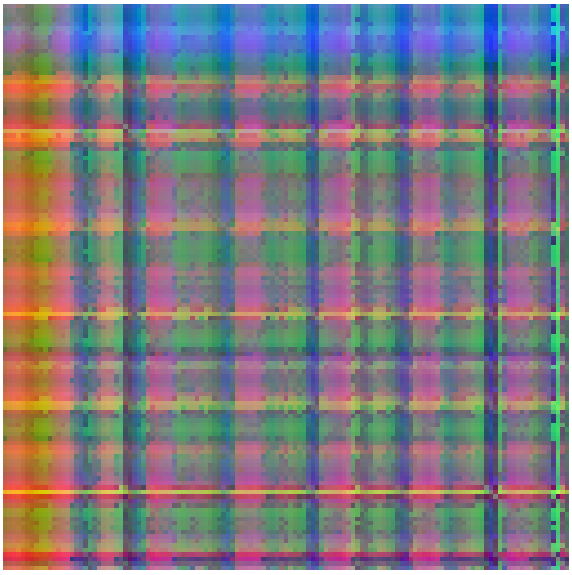


Figure 2: Example of a Recurrence Plot.

The adjacent-state construction produces $N - 1$ embedded states rather than N scalar samples. The information loss is not merely a count mismatch, since the overlapping states still contain all samples in principle; it arises because the RP stores pairwise distances or thresholded recurrences, which are invariant to transformations such as translation, rotation, reflection, and, depending on normalization, scale. The signed Lu–Tong convention used here adds an orientation cue, but it does not make the RP representation fully bijective.

2.1.2. Time Series Reconstruction

There is no direct inverse transformation for converting time series into recurrence plots according to the literature. However, several approaches have been proposed over time, primarily because this is one of the oldest methods for reconstruction. Notable contributions include those by M. Thiel [5] and Tanio et al. [6]. In particular, we highlight the decoding method by Hirata, Horai, and Aihara [7], which reproduces the original time series from a recurrence plot by feeding graph geodesics into multidimensional scaling and applies to both continuous and binary (unsigned) recurrence plots; we describe it here as the closest precedent and retain it as the prior-art baseline of Section 6.6. Because its multidimensional-scaling embedding is fixed only up to an orthogonal transformation, that reconstruction is recovered only up to a reflection — an ambiguity the authors themselves note. Our own decoder (Section 4.2) targets the specific case their method does not address, reconstructing the scalar sequence from the *continuous signed* Lu–Tong RP of overlapping states: it departs from the graph-geodesic construction — recovering the Euclidean

distances directly from the corrected inverse normalisation and applying classical MDS to them — and adopts the signed Lu–Tong definition (Equation 4) as an in-loop orientation constraint that resolves the reflection their setting leaves open.

1. **Starting from a Recurrence Plot:** Let $z(i)$ denote the state at time i (with $1 \leq i \leq n$). Define a function c that takes two states and indicates whether they are sufficiently close; it returns 1 if they are close and 0 otherwise, according to a predefined threshold. Thus, the recurrence plot can be represented as a binary cost matrix:
$$\{(i,j) : c(z_i, z_j) = 1, \quad i,j = 1,2,\ldots,n\}.$$

2. **Graph Construction:** Construct a graph in which an edge is drawn between nodes i and j if $c(z_i, z_j) = 1$.

3. **Edge Weight Assignment:** For each state z_i, define the set
$$G_i = \{j : c(z_i, z_j) = 1\},$$
which represents the indices of time where state z_i is close to other states. Then, assign a weight to each edge between i and j as follows:
$$w(i,j) = 1 - \frac{|G_i \cap G_j|}{|G_i \cup G_j|}.$$

4. **Shortest Path Calculation:** Compute the shortest path distances in the graph using Dijkstra's algorithm [8]. For larger graphs, an alternative method for finding the shortest path can be employed.

5. **Multidimensional Scaling:** Finally, apply a multidimensional scaling technique that orders the set of points using the component corresponding to the largest eigenvalue. We use *classical* (Torgerson) MDS, i.e. the closed-form double-centering eigendecomposition of $-\frac{1}{2}JD^2J$ with $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, rather than the SMACOF iterative optimiser. For an exactly Euclidean distance matrix, classical MDS gives an exact embedding when the required positive-eigenvalue dimensions are retained; our decoder (Section 4.2) keeps the two leading positive-eigenvalue coordinates to recover the full adjacent-state geometry, which classical MDS produces deterministically without the stochastic initialisation dependence of an iterative SMACOF solver.

Although this graph-based method can recover a consistent state ordering, it has a notable drawback, also discussed in [7]: multidimensional scaling may return a mirror-symmetric reconstruction. In our WISDM experiments, this ambiguity was observed before applying the signed-RP orientation check described below; therefore, the observed ambiguity should be understood as an empirical diagnostic of this implementation rather than as a parameter from the cited method.

2.2. Gramian Angular Field

2.2.1. Image Construction

Introduced by Wang, Z. and Oates, T. in *Imaging Time-Series to Improve Classification and Imputation* [9], the Gramian Angular Field (GAF) technique represents time series values in polar coordinates. To do so, it is essential to rescale the time series values to an appropriate interval either to $[-1, 1]$ to map to angles in $[0, \pi]$, or to $[0, 1]$ to map to angles in $[0, \pi/2]$.

Once the time series is rescaled, the encoding is defined using the arccosine and the radius. Specifically, for the interval $[0, \pi]$ the definitions are:

$$\begin{cases} \phi = \arccos(\tilde{x}_i), & -1 \leq \tilde{x}_i \leq 1, \quad \tilde{x}_i \in \tilde{X}, \\ r = \frac{t_i}{N}, & t_i \in \mathbb{N}, \end{cases} \tag{5}$$

and for the interval $[0, \pi/2]$ they become:

$$\begin{cases} \phi = \arccos(\tilde{x}_i), & 0 \leq \tilde{x}_i \leq 1, \quad \tilde{x}_i \in \tilde{X}, \\ r = \frac{t_i}{N}, & t_i \in \mathbb{N}. \end{cases} \tag{6}$$

According to Wang and Oates [9], the polar transformation preserves temporal ordering through the radius and maps normalized amplitudes to angles through a monotonic arccosine branch. This does not mean that every GAF matrix has the same inverse properties. GASF and GADF use different pairwise angular operators, so the reconstruction argument must be derived separately for each case: GASF exposes amplitude information on the diagonal, whereas GADF stores only relative angular differences and has a zero diagonal.

Finally, two approaches to obtain a GAF are defined:

$$\begin{aligned} G_{ij}^{\mathrm{S}} &= \cos(\phi_i + \phi_j) \\ &= \tilde{x}_i \tilde{x}_j - \sqrt{1-\tilde{x}_i^2}\sqrt{1-\tilde{x}_j^2}, \end{aligned} \tag{7}$$

$$\begin{aligned} G_{ij}^{\mathrm{D}} &= \sin(\phi_i - \phi_j) \\ &= \sqrt{1-\tilde{x}_i^2}\tilde{x}_j - \tilde{x}_i\sqrt{1-\tilde{x}_j^2}. \end{aligned} \tag{8}$$

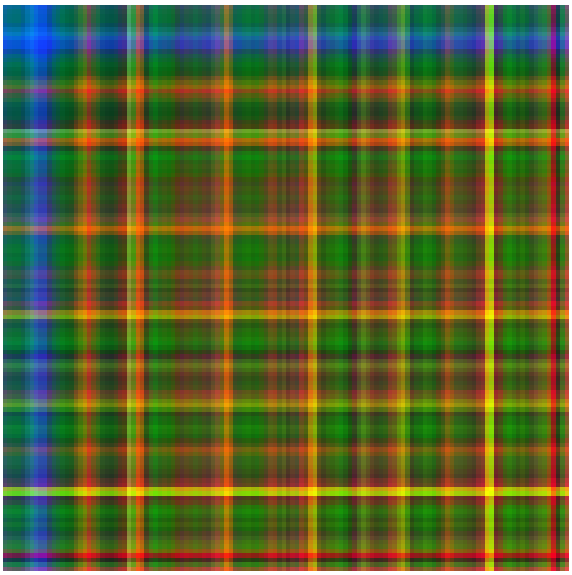


Figure 3: Example of a Gramian Angular Field.

2.2.2. Time Series Reconstruction

The two GAF variants require different reconstruction arguments. GASF has a known diagonal-based inverse under the appropriate normalization range, whereas GADF cannot be treated by simply reusing the same diagonal argument. We therefore separate the current GASF baseline from the GADF off-diagonal reconstruction developed below.

GASF reconstruction. In the original GAF paper [9], the possibility of recovering the original time series is connected to the structure of the Gramian representation. In subsequent literature, including a later paper by the same authors [10], the practical reconstruction route is discussed through the Gramian Angular Summation Field (GASF) main diagonal.

For GASF inversion, we use the main-diagonal argument. The diagonal half-angle reconstruction is already noted in the original GAF paper by Wang and Oates [9]; we follow the explicit normalization conditions later stated by Tao, Deng, and Xiao [11], under which the diagonal can be used for reconstruction. Under those conditions, the normalized time series can be reconstructed by observing the main diagonal of a GASF. Specifically, since the diagonal elements satisfy

$$G_{ii}^{\mathrm{S}} = \cos(2\phi_i),$$

one can recover the normalized sample using the half-angle identity:

$$\tilde{x}_i = \cos\phi_i = \sqrt{\frac{G_{ii}^{\mathrm{S}} + 1}{2}}, \quad \phi_i \in \left[0, \frac{\pi}{2}\right].$$

It is important to note that the main-diagonal inversion is not unique for the interval $[-1, 1]$, which maps to angles in $[0, \pi]$. In that case,

$$G_{ii}^{\mathrm{S}} = \cos(2\phi_i) = 2\tilde{x}_i^2 - 1,$$

so the half-angle identity recovers only $|\tilde{x}_i|$, not the sign of $\tilde{x}_i$. This limitation applies to the diagonal-only inversion rule used here; it does not constitute a proof that no reconstruction could be obtained from the full GASF matrix or from additional side information.

Wang and Oates state that the GADF of $[0, 1]$-rescaled data has an accurate inverse map [9]. However, their reconstruction discussion develops the practical inverse through the GASF diagonal and does not provide an explicit

GADF decoding algorithm. In particular, a GADF decoder must resolve the branch of arcsin, fix the otherwise free global angular offset implied by the difference-only form $\sin(\phi_i - \phi_j)$, keep the sign convention fixed, and map the recovered angular trajectory back to normalised amplitudes; recovery of physical accelerometer units is an optional external calibration step. Reconstruction via the main diagonal is not feasible since the angular differences yield

$$G_{ii}^{\mathrm{D}} = \sin(\phi_i - \phi_i) = 0,$$

regardless of whether the time series is scaled to $[-1, 1]$ with angles in $[0, \pi]$ or to $[0, 1]$ with angles in $[0, \pi/2]$. The method proposed later in Section 4.4 therefore uses off-diagonal angular differences rather than the diagonal.

2.3. Markov Transition Field

2.3.1. Image Construction

Introduced by Wang and Oates in *Imaging Time-Series to Improve Classification and Imputation* [9], the Markov Transition Field (MTF) is probabilistic rather than geometric or trigonometric. Given a time series X, the values are divided into Q quantile bins and each sample is assigned a discrete state $q_t \in \{1, \ldots, Q\}$. The first-order Markov transition matrix is estimated as

$$W_{p,q} = P(q_{t+1} = q \mid q_t = p) = \frac{\#\{t : q_t = p,\ q_{t+1} = q\}}{\#\{t : q_t = p\}},$$

with rows normalized to sum to one when the source state is observed.

The MTF expands this $Q \times Q$ matrix back to an $N \times N$ image by assigning each pixel the transition probability associated with the states of its two time indices:

$$M_{a,b} = W_{q_a, q_b}, \qquad a, b = 1, \ldots, N. \tag{9}$$

Thus, each pixel stores a transition statistic between quantile states rather than a direct distance or angular relation between amplitudes. This explains why MTF can be robust for classification while being weak for pointwise reconstruction: all samples inside the same quantile share the same reconstructed representative value.

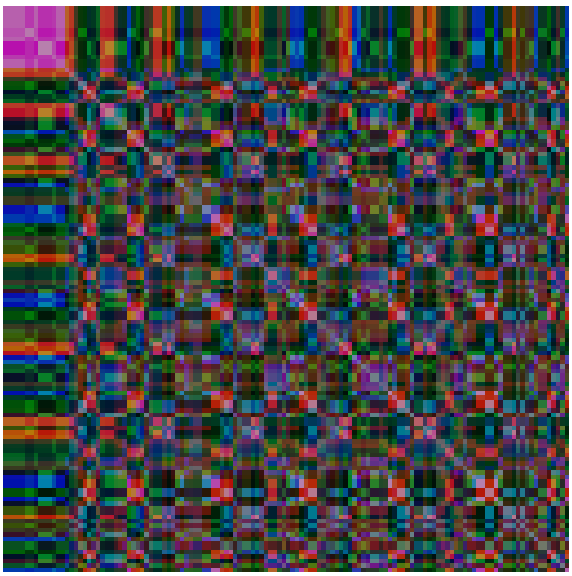


Figure 4: Example of a Markov Transition Field.

2.3.2. Time Series Reconstruction

In the literature, there is no documented method for estimating the original time series from a Markov Transition Field (MTF) matrix. Some articles use Markov Random Fields for image estimation and reconstruction, applying them as image filters to improve reconstructed image quality [12, 13]. However, those methods address image restoration rather than inverse MTF decoding.

3. Problem Formulation

Let $x = (x_1, \ldots, x_N)$ denote one axis of a triaxial accelerometer window, with $N = 128$ in all experiments. Each method m first maps the raw window into its own normalised domain $\mathcal{X}_m$ through a per-window, method-specific normaliser $\mathcal{N}_m$ (formalised in Section 4.1), $\tilde{x}^{(m)} = \mathcal{N}_m(x)$; the image construction then acts on that normalised signal,

$$\mathcal{E}_m : \mathcal{X}_m \to \mathbb{R}^{N_m \times N_m},$$

a single-channel matrix per accelerometer axis. In our setting, $N_m = N$ for GASF, GADF, and MTF, while the modified RP representation internally uses adjacent states and therefore has $N_m = N - 1$ before the recovered adjacent states are mapped back to N samples through their overlap. The inverse problem studied in this paper is to construct a decoder

$$\mathcal{D}_m : \mathbb{R}^{N_m \times N_m} \to \mathcal{X}_m$$

that returns a *normalised* reconstruction $\hat{\tilde{x}}^{(m)} = \mathcal{D}_m(\mathcal{E}_m(\tilde{x}^{(m)}))$ close to $\tilde{x}^{(m)}$ under pointwise, trend-level, and downstream classification criteria; recovery of the physical-scale window $\hat{x} = \mathcal{N}_m^{-1}(\hat{\tilde{x}}^{(m)}) \in \mathbb{R}^N$ is the optional final step of the shared protocol. A triaxial window is encoded axis by axis; the three single-channel matrices are stacked into one $N_m \times N_m \times 3$ RGB image for storage, and the decoder is applied independently to each channel before the recovered axes are regrouped into a single window.

This formulation separates three sources of difficulty. First, some encodings introduce invariances or many-to-one summaries by construction. In RP, the adjacent-state embedding produces $N - 1$ overlapping two-dimensional states that still contain all samples in principle, but the stored pairwise distances or recurrences are invariant to transformations such as translation, rotation, reflection, and, depending on normalization, scale; the decoder then also has to map the recovered $N - 1$ state sequence back to N samples. Second, some encodings introduce branch or orientation ambiguities, as in GASF diagonal inversion, GADF sine-difference inversion, and multidimensional scaling for RP. Third, image storage and normalization introduce practical quantisation and rescaling errors. The proposed decoders therefore report their assumptions explicitly: normalization range, required encoder settings or calibration, orientation convention, and whether the reconstruction is exact, approximate, or heuristic.

The rest of the paper follows this formulation. Section 4 describes the decoders, including the proposed GADF multi-pivot, boundary-anchored reconstruction. Section 5 defines the data and metrics, Section 6 reports the complete empirical evidence for all four encodings, and Section 7 interprets the trade-offs among encodings. This organization is intentional: methods may be mathematically defined before all empirical evidence is available, but comparative claims are restricted to the decoders tested under the same protocol.

4. Proposed Reconstruction Methods

This section describes the reconstruction strategy associated with each encoding. Each method is presented as an inverse map with explicit assumptions, so that later experiments can be read as tests of these assumptions rather than as isolated implementation results.

4.1. Shared normalisation, storage, and calibration protocol

All four encoders and decoders operate on a per-window, per-axis normalised signal, and the mapping back to physical units is a single optional external stage shared by every method (the normalisation and optional physical-scale-recovery blocks in Fig. 1). Writing $\mathcal{N}_m$ for the per-window, method-specific normaliser and $\mathcal{E}_m, \mathcal{D}_m$ for the encoder and decoder of method m, the pipeline is

$$\tilde{x} = \mathcal{N}_m(x), \qquad \hat{\tilde{x}} = \mathcal{D}_m\big(\mathcal{E}_m(\tilde{x})\big),$$

where the encoded matrix is stored and re-read as an 8-bit PNG between $\mathcal{E}_m$ and $\mathcal{D}_m$: each matrix M is quantised as $\mathrm{round}(255\, g_m(M))$ into a `uint8`, with the three axes stored as the R/G/B channels of one image; g_m min–max-normalises per image for RP and the two Gramian fields and is the identity on the fixed $[0, 1]$ range for MTF, and the decoder applies the corresponding inverse ($[0, 255] \to [0, 1]$ for MTF, $\to [-1, 1]$ for the Gramian fields, and $\to$ the stored per-axis *RP-matrix* range $[R_{\min}, R_{\max}]$ for RP). MTF is therefore the only store/read pair whose intensity range is fixed *a priori*, so reading it back needs no stored per-image range — a statement about the PNG round-trip only, *not* about the MTF$\to$series decode, which stays many-to-one. For GASF the encoded matrix already occupies the full $[-1, 1]$ range — both extremes are attained under per-window normalisation — so the fixed $[-1, 1]$ read inverts the per-image min–max exactly. For GADF the entries occupy the marginally narrower $[-\beta, \beta]$, $\beta = 2\alpha\sqrt{1-\alpha^2} \approx 0.99980$ at $\alpha = 0.7$ (Section 4.4), so decoding on the fixed $[-1, 1]$ range rescales each recovered entry by $1/\beta \approx 1.0002$; this bounded mismatch is at most $\approx 2 \times 10^{-4}$ per entry — some 40$\times$ below the $2/255 \approx 7.8 \times 10^{-3}$ eight-bit quantisation step — and, because the downstream arcsin reconstruction is dominated by that quantisation, decoding on the exact

$[-\beta, \beta]$ range in place of $[-1, 1]$ does not measurably change the reconstruction. For RP the read requires the stored per-window RP-matrix range $[R_{\min}, R_{\max}]$: inverting with that range — rather than the signal amplitude range — recovers the signed distances themselves and is what makes the recovered matrix a valid Euclidean distance matrix (Section 4.2). **Unless otherwise stated, every decoder returns its reconstruction in the normalised domain** $\hat{\tilde{x}}$; the per-axis minimum and maximum $(x_{\min}, x_{\max})$ recorded at encoding time are used *only* when the normalised series must be mapped back to the physical accelerometer units,

$$\hat{x} = \mathcal{N}_m^{-1}(\hat{\tilde{x}};\, x_{\min}, x_{\max}).$$

Metadata dependence is therefore conditional on both the encoding and the desired output domain: the Gramian and MTF decoders need no per-window metadata for normalised reconstruction (only the image and the known encoder settings), RP additionally needs the per-window RP-matrix range $[R_{\min}, R_{\max}]$ for its normalised reconstruction, and physical-scale reconstruction of any encoding additionally requires the original per-axis amplitude calibration $(x_{\min}, x_{\max})$. The *normalisation range*, by contrast, is part of each encoder–decoder definition and is not interchangeable: the diagonal GASF inverse assumes the $[0, 1]$ range (angles in $[0, \pi/2]$), the restricted-branch GADF inverse assumes $[-\alpha, \alpha]$ with $\alpha < 1/\sqrt{2}$, MTF discretises the normalised range into Q quantile bins, and the signed-RP orientation check requires the encoder and decoder to share the same signed-RP normalisation convention. Each subsection below therefore states its normalisation range explicitly, returns a normalised reconstruction, and defers the optional physical-scale step to the common $\mathcal{N}_m^{-1}$ above.

4.2. Recurrence Plot

The Recurrence Plot decoder proposed below is a *new* contribution of this work; the closest precedent in the literature is the reconstruction of Hirata, Horai, and Aihara [7], which reproduces the original time series from an *unsigned* recurrence plot through graph-geodesic multidimensional scaling, but whose embedding is fixed only up to an orthogonal transformation and therefore leaves a reflection ambiguity; it does not address the *continuous signed* RP of overlapping states treated here. Our decoder closes that specific gap by exploiting a property specific to the signed Lu–Tong RP under the L_2 norm: each off-diagonal entry is (up to sign) the Euclidean distance between the two-dimensional adjacent states $s_i = (x_i, x_{i+1})$, so the recovered magnitude matrix is a genuine Euclidean distance matrix between those states. We therefore (i) invert the stored image back to the RP's *own* distance range rather than the signal amplitude range, (ii) recover the full two-dimensional state cloud with classical multidimensional scaling, (iii) fix the residual orthogonal transformation with an overlap constraint between consecutive states and the signed-RP convention, and (iv) read the scalar series directly from the overlapping coordinates. Correcting the inverse normalisation is what makes the recovered matrix a valid distance matrix; the earlier one-dimensional projection reported in preliminary experiments decoded the image with the signal amplitude range, which affinely distorts the distances handed to multidimensional scaling and collapses the two-dimensional geometry. With the correction the encoder–decoder pair is near-invertible on the protocol of this paper (Pearson ≈ 0.99998, mean absolute error ≈ 0.015; Table 18), on par with the Gramian fields.

Multidimensional scaling recovers the two-dimensional state cloud only up to a global orthogonal transformation and translation; the decoder therefore aligns that cloud with the overlap structure of consecutive adjacent states and resolves the residual reflection with the signed-RP convention. Concretely:

- **Step 1 (corrected inverse normalisation):** Recover the signed RP matrix R^{obs} from the decoded image by inverting the PNG quantisation with the RP matrix's *own* stored range $[R_{\min}, R_{\max}]$ — the per-window, per-axis minimum and maximum of the signed RP — and *not* the signal amplitude range. This is the decisive correction: because the encoder min–max-normalises each RP by its own range, inverting with the signal range applies an affine map $R^{\text{obs}} = aR + b$ with $b \neq 0$, which shifts the zero of the distance matrix and destroys its Euclidean structure. The RP range is one extra scalar pair per axis — six scalars per triaxial window — stored alongside the signal range already recorded for physical-scale recovery.
- **Step 2 (magnitude matrix):** Form the symmetric magnitude matrix $D_{ij} = \frac{1}{2}\big(|R^{\text{obs}}_{ij}| + |R^{\text{obs}}_{ji}|\big)$ with $D_{ii} = 0$; the symmetrisation absorbs the small asymmetries introduced by PNG quantisation. Under the L_2 signed-RP definition, $D_{ij} = \|s_i - s_j\|_2$ is the Euclidean distance between the adjacent states $s_i = (x_i, x_{i+1})$.
- **Step 3 (two-dimensional embedding):** Apply classical (Torgerson) multidimensional scaling to D and retain the two leading positive-eigenvalue coordinates, giving $Y = V_2 \Lambda_2^{1/2} \in \mathbb{R}^{(N-1)\times 2}$. For a noise-free Euclidean D this recovers the state cloud exactly, up to an orthogonal transformation $Q \in O(2)$ and a translation.
- **Step 4 (overlap alignment):** The true states satisfy the overlap identity $s_{i,2} = s_{i+1,1} = x_{i+1}$. We search the orthogonal group for the transform that best enforces it: for each rotation angle θ and reflection parity $\eta \in \{\pm 1\}$ we form $Q_{\theta,\eta}$ and $Z = YQ_{\theta,\eta}$, correct the translation offset $\delta = -\frac{1}{N-2}\sum_i (Z_{i,2} - Z_{i+1,1})$ via $a_i = Z_{i,1} - \delta/2$, $b_i = Z_{i,2} + \delta/2$, and score the orientation by the normalised overlap error $E_{\text{overlap}} = \sum_i (b_i - a_{i+1})^2 \big/ \big(\sum_i (a_i^2 + b_i^2) + \epsilon\big)$.

- **Step 5 (signed orientation selection):** Among the lowest-overlap-error orientations, we regenerate the signed RP of each candidate reconstruction using Equation 4 and keep the one whose signed RP is closest to R^{obs} in Frobenius norm. The signed convention thus resolves the remaining rotation and reflection, generalising the earlier binary sign check that chose only between a reconstruction and its global negation.
- **Step 6 (overlap reconstruction):** From the aligned states $\hat{s}_i = (a_i, b_i)$ the scalar series follows directly from the overlap, recovering all N samples from the $N-1$ states without any appended constant:

$$\hat{x}_1 = a_1, \qquad \hat{x}_i = \tfrac{1}{2}\big(b_{i-1} + a_i\big) \;\; (2 \le i \le N-1), \qquad \hat{x}_N = b_{N-1}.$$

  Averaging the two estimates of each shared sample also attenuates residual multidimensional-scaling and quantisation noise.
- **Step 7 (amplitude anchoring):** The overlap reconstruction of Step 6 is determined only up to a global additive constant: both the pairwise distances and the Lu–Tong signs are invariant to a common offset added to the whole window, so $\hat{x}_i$ and $\hat{x}_i + c$ regenerate the same signed RP. The decoder fixes this constant by min–max mapping the recovered sequence's own $[\min, \max]$ onto the known target amplitude range — the RP encoder's per-window normalised range for the normalised reconstruction, or $[x_{\min}, x_{\max}]$ for physical accelerometer units. This is well defined because per-window min–max normalisation forces the window to attain both extremes, so the recovered minimum and maximum coincide with the true endpoints of the range; the same map restores the amplitude origin and, when the physical range is chosen, the physical scale, coinciding there with the shared calibration $\mathcal{N}_m^{-1}$ of Section 4.1.

With the corrected inverse normalisation the recovered D is, up to quantisation, a valid two-dimensional Euclidean distance matrix: multidimensional scaling explains essentially all of its variance in two dimensions and the overlap reconstruction returns the signal almost exactly (Table 18). Two algorithmic choices are decisive once the image is decoded on its own RP distance range: retaining the full two-dimensional state geometry and resolving its orientation through the signed RP. Decoding on the wrong, signal-amplitude range is a round-trip error rather than a method variant — it affinely distorts the distances and collapses the two-dimensional geometry — while without the signed orientation the reflection remains unresolved.

4.3. Gramian Angular Summation Field Reconstruction

Unlike the other three decoders in this section, the GASF inverse is *not* a new contribution of this work: it follows the diagonal-inverse strategy used by Huamin et al. [11] in the 2-D representation-based denoising autoencoder line of work. We re-derive it here in the notation of the rest of the paper and re-implement it inside the same pipeline used by the other three decoders, so the GASF numbers reported later can be read as a tightly controlled prior-art baseline against which the three novel decoders (RP, GADF, MTF) are compared. The contribution of this subsection is therefore the explicit normalisation and quantisation handling that make the inverse stable when the GASF matrix is read back from an 8-bit PNG, not the inverse map itself.

The GASF decoder is the most direct inverse considered in the completed experiments because its main diagonal contains a sample-wise angular constraint. Under the branch used in this work, the encoded values are first interpreted through

$$G^{\text{S}}_{ii} = \cos(2\phi_i),$$

and the angle is recovered as

$$\hat{\phi}_i = \frac{1}{2}\arccos\left(G^{\text{S}}_{ii}\right).$$

The normalized signal is then obtained by

$$\hat{\tilde{x}}_i = \cos(\hat{\phi}_i).$$

This rule is exact only when the normalization and angular branch are consistent with the encoder. If the full interval $[-1, 1]$ is used without additional sign information, the diagonal determines $|\tilde{x}_i|$ rather than $\tilde{x}_i$ itself. Therefore, the high numerical accuracy reported for GASF should be read as the result of a controlled normalization protocol, not as an unconditional inverse for every possible GAF image.

The recovered $\hat{\tilde{x}}_i = \cos(\hat{\phi}_i)$ is the normalised reconstruction; when physical accelerometer units are wanted, it is passed through the common calibration $\mathcal{N}_m^{-1}$ of Section 4.1. The normalised shape is thus recovered from the diagonal alone, and the recorded per-window min–max is needed only for the optional physical-scale step — which is also why a generated GASF image would require plausible scale metadata only if physical units, rather than normalised shape, are required.

4.4. Restricted-Branch Gramian Angular Difference Field Reconstruction

We now describe a restricted-branch GADF decoder. Wang and Oates note that GADF can admit an accurate inverse map when the data are rescaled to $[0, 1]$ [9], but their reconstruction discussion operationalises the inverse through the GASF diagonal and does not specify a GADF decoding algorithm. In particular, GADF stores $\sin(\phi_i - \phi_j)$, so an operational inverse must choose an arcsin branch, fix the otherwise free global angular offset, keep a sign convention fixed, and map the recovered angular trajectory back to normalised amplitudes, with physical-unit recovery left as an optional external calibration step. The contribution here is therefore not the abstract observation that GADF may be invertible under a restricted angular range, but a fully specified operational decoder: a closed-form multi-pivot, boundary-anchored reconstruction with an explicit branch condition, an image-derived angular anchor, a fixed sign convention, optional min–max calibration for physical-scale recovery, and empirical evaluation under the same protocol as RP, MTF, and GASF. It should not be read as a generic inverse for every standard GADF image on the full $[-1, 1]$ range. The novelty lies in two coupled design choices: (i) we shrink the normalised amplitude range to $[-\alpha, \alpha]$ with $\alpha = 0.7$ so that *every* pairwise angular difference is forced inside the principal branch of arcsin, which is what turns the inverse from an iterative optimisation into a one-pass closed-form computation, and (ii) we anchor the otherwise-relative angular trajectory by reading the angular boundary directly off the extreme row of the GADF image (rather than from an external time-domain index), so that the normalised trajectory is recovered from the image alone — the stored $(x_{\min}, x_{\max})$ being needed only for optional physical-scale recovery.

GADF requires a reconstruction strategy different from the GASF diagonal inverse. The diagonal of G^{D} is identically zero, but the full matrix stores relative angular information through $G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j)$. The proposed decoder uses this off-diagonal structure and turns a row of the GADF matrix into a set of recoverable angular differences.

4.4.1. Restricted scaling for branch uniqueness

The general GADF inverse is not unique without additional assumptions because $\sin(\theta) = \sin(\pi - \theta)$. The key design decision is therefore to restrict the normalized amplitude range before encoding. This sacrifices part of the nominal $[-1, 1]$ GAF range, but it keeps all pairwise angular differences inside the principal branch of arcsin, which is what makes a closed-form decoder possible. The input series is scaled as

$$\bar{x}_t = 1.4\frac{x_t - x_{\min}}{x_{\max} - x_{\min}} - 0.7, \qquad \bar{x}_t \in [-\alpha, \alpha], \quad \alpha = 0.7.$$

The corresponding angles satisfy

$$\phi_t = \arccos(\bar{x}_t) \in [\arccos(\alpha), \arccos(-\alpha)] \approx [0.795, 2.346].$$

The value $\alpha = 0.7$ is not an arbitrary hyperparameter. It is chosen as a near-maximal symmetric amplitude range that preserves the injectivity of the inverse-sine step. For the closed-form GADF inverse to be unique, every angular difference must remain inside the principal branch of arcsin:

$$|\phi_i - \phi_j| < \frac{\pi}{2}.$$

The worst-case angular spread is

$$\arccos(-\alpha) - \arccos(\alpha).$$

Using $\arccos(-\alpha) = \pi - \arccos(\alpha)$, the uniqueness condition becomes

$$\pi - 2\arccos(\alpha) < \frac{\pi}{2},$$

or equivalently

$$\alpha < \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \approx 0.7071.$$

We therefore set $\alpha = 0.7$, close to the largest admissible symmetric range while leaving a small numerical margin:

$$|\phi_i - \phi_j| \le \arccos(-0.7) - \arccos(0.7) \approx 1.551 < \frac{\pi}{2}.$$

This keeps every angular difference within the principal branch of arcsin. Under this restriction,

$$\arcsin\left(G^{\mathrm{D}}_{ij}\right) = \phi_i - \phi_j.$$

Consequently the encoded GADF entries do not fill the nominal $[-1, 1]$ GAF range but occupy the slightly narrower interval $[-\beta, \beta]$, whose bound is attained between the two extreme samples:

$$\beta = \sin\big(\arccos(-\alpha) - \arccos\alpha\big) = \sin\big(2\arccos\alpha\big) = 2\alpha\sqrt{1 - \alpha^2},$$

which equals 0.99980 at $\alpha = 0.7$. Per-window normalisation forces both amplitude extremes to be present, so this bound is actually reached in every non-constant window and the encoder's per-image min–max maps exactly $[-\beta, \beta] \to [0, 1]$. The residual gap between $[-\beta, \beta]$ and the fixed $[-1, 1]$ storage range therefore matters only for the eight-bit round-trip, where it is a bounded sub-quantisation effect (Section 4.1). Without the restriction $\alpha < 1/\sqrt{2}$, the same value $\sin(\phi_i - \phi_j)$ could correspond to multiple angular-difference configurations, and the GADF matrix would no longer determine the angular differences uniquely. The branch restriction gives a single angular reconstruction under the controlled protocol, once the absolute angular anchor is also fixed (the min–max metadata is not needed for this normalised reconstruction, only for the optional physical-scale recovery). The reliability check of Section 6.5 confirms at $\alpha = 0.7$ that the floating-point inverse is exact, separates it from 8-bit PNG quantisation error, and finds a 0% anchor-misidentification rate on WISDM.

4.4.2. Relative angular reconstruction from a pivot row

For a pivot row p, the GADF entries provide

$$G^{\mathrm{D}}_{pj} = \sin(\phi_p - \phi_j),$$

and, by the branch condition above,

$$\phi_j = \phi_p - \arcsin\left(G^{\mathrm{D}}_{pj}\right), \qquad j = 1, \ldots, N.$$

Thus a pivot row determines the entire angular trajectory only up to a global offset; the absolute offset is fixed separately by the boundary (extreme-mean) anchor described below. The pivot is chosen from the image itself rather than from an external time-domain index, because a stored or generated GADF image may not preserve the original contextual information. We select the pivot row as

$$p = \arg\min_i \left| \frac{1}{N} \sum_{j=1}^{N} G^{\mathrm{D}}_{ij} \right|,$$

which chooses the row whose average signed angular difference is closest to zero. Because the row mean equals $R\sin(\phi_i - \delta)$ (derived below), this selects rows whose angle ϕ_i is close to the circular-mean angle δ of the window — *not*, in general, to $\pi/2$: if the amplitudes concentrate near one extreme, δ can drift toward that boundary even though per-window normalisation guarantees both extremes are present. In practice the decoder does not rely on a single pivot: it averages the reconstruction over the $n_p = 5$ most-central rows (the five smallest |row mean|), which cancels most of the per-row 8-bit quantisation noise (Section 6.5 quantifies the gain). Selecting central pivots *tends* to keep the recovered differences $G^{\mathrm{D}}_{pj} = \sin(\phi_p - \phi_j)$ small, so most of the arcsin arguments the decoder reads fall in its near-linear regime where 8-bit quantisation is least amplified. This is an *empirical* tendency rather than a guarantee: on WISDM the selected pivots sit close to the angular centre (mean $|\phi_p - \frac{\pi}{2}| \approx 0.13$ rad) and the pivot-row entries are small on average (mean $|G^{\mathrm{D}}_{pj}| = 0.22$, 99th percentile 0.76), but the worst case can still approach the branch boundary (max ≈ 0.99) for windows whose amplitude distribution is strongly skewed within the admissible interval.

4.4.3. Absolute angular anchor

The pivot row reconstructs angles relative to ϕ_p, so one absolute angular anchor is still required — but this anchor is supplied by the *known* normalisation range, not by the stored min–max. Because the normalisation is performed per window, the encoded normalised signal necessarily reaches the two boundary values $+\alpha$ and $-\alpha$ unless the window is constant, so the extreme samples sit at fixed, *a priori* known angular boundaries:

$$\phi_{\max} = \arccos(\alpha), \qquad \phi_{\min} = \arccos(-\alpha).$$

Operationally, the boundary row is identified from the image as the extreme row

$$e = \arg\max_i \left| \frac{1}{N} \sum_{j=1}^{N} G^{\mathrm{D}}_{ij} \right|.$$

The sign of the mean row indicates which boundary is represented: a positive mean corresponds to the large-angle boundary $\phi_e = \arccos(-\alpha) = \phi_{\min}$, whereas a negative mean corresponds to the small-angle boundary $\phi_e = \arccos(\alpha) = \phi_{\max}$.

This identification is exact in the noise-free restricted regime, because the row mean is a strictly monotonic function of the row angle. Writing $\bar{C} = \frac{1}{N}\sum_j \cos\phi_j$ and $\bar{S} = \frac{1}{N}\sum_j \sin\phi_j$,

$$\frac{1}{N}\sum_{j=1}^{N} G^{\mathrm{D}}_{ij} = \bar{C}\sin\phi_i - \bar{S}\cos\phi_i = R\,\sin(\phi_i - \delta), \qquad R = \sqrt{\bar{C}^2 + \bar{S}^2}, \quad \delta = \operatorname{atan2}(\bar{S}, \bar{C}),$$

whose derivative $R\cos(\phi_i-\delta)$ is strictly positive over the whole admissible range. Indeed every $\phi_j \in [\arccos\alpha, \arccos(-\alpha)]$, and δ is the argument of a convex combination of the unit vectors $(\cos\phi_j, \sin\phi_j)$, so δ lies in that same interval and $\phi_i - \delta \in [-(\arccos(-\alpha) - \arccos\alpha), \arccos(-\alpha) - \arccos\alpha]$. For $\alpha = 0.7$ this spread is $\approx 1.551 < \pi/2$ — the same margin that keeps the arcsin branch unique — hence $\cos(\phi_i - \delta) > 0$ throughout and the row mean is strictly increasing in ϕ_i. Its extreme value is therefore attained exactly at the boundary sample ($\pm\alpha$) that the per-window normalization is guaranteed to produce, and the sign of that extreme selects the correct boundary. The $\alpha = 0.7$ restriction thus secures branch uniqueness and unambiguous anchor identification at once.

Once a boundary row is identified, it fixes the otherwise-free global offset through a *single* entry per pivot: for each central pivot p,

$$\phi_p = \phi_e - \arcsin\left(G^{\mathrm{D}}_{ep}\right),$$

after which the full angular trajectory follows from that pivot row — not from the extreme row — through $\phi_j = \phi_p - \arcsin(G^{\mathrm{D}}_{pj})$ of the previous subsection. The decoder does not trust a single pivot: it averages the recovered *angles* over the set P of n_p central rows and only then maps back to amplitude,

$$\hat{\phi}^{(p)}_j = \phi_e - \arcsin\big(G^{\mathrm{D}}_{ep}\big) - \arcsin\big(G^{\mathrm{D}}_{pj}\big), \qquad \hat{\phi}_j = \frac{1}{n_p}\sum_{p\in P}\hat{\phi}^{(p)}_j, \qquad \hat{\tilde{x}}_j = \cos\Big(\hat{\phi}_j\Big), \quad j = 1, \ldots, N.$$

Averaging the *angles* — not the reconstructed amplitudes — is what the implementation does, and it is what lowers the reconstruction error as n_p grows (Section 6.5). The reconstruction is unique within the controlled protocol: the central pivot rows supply the trajectory shape, while the boundary value imposed by the per-window normalisation — read through the anchor scalar G^{D}_{ep} (empirically ≈ 0.78 on WISDM: moderate, though, unlike the 0.70 of an exactly-central pivot, not pinned away from the branch edge) — fixes the global offset. If the stored matrix uses the transposed GADF convention $\sin(\phi_j - \phi_i)$, every arcsin sign above flips; the implementation must therefore keep the convention in Equation 8 fixed.

4.4.4. Normalised output and limitations

The GADF image, the fixed value of α, and the encoding sign convention are sufficient to recover the normalised trajectory $\hat{\tilde{x}}_j = \cos(\phi_j) \in [-\alpha, \alpha]$ directly from the image; the per-window minimum and maximum are *not* required for this step. The calibration pair $(x_{\min}, x_{\max})$ enters only through the common physical-scale map $\mathcal{N}_m^{-1}$ of Section 4.1, i.e. only when the reconstructed series must be expressed in the original accelerometer units; metadata dependence is thus conditional on the output domain. The method is closed-form but not linear-time overall: selecting the anchor and pivot rows requires the row-mean statistics over the full $N \times N$ input matrix, so the decoder has $O(N^2)$ total cost — dominated by those row statistics (equivalently, by reading the N^2 stored pixels) — followed by an $O(N)$ closed-form angular reconstruction from each selected row once the anchor is fixed. It is also not universally invertible: it relies on the restricted scaling interval $[-\alpha, \alpha]$ with $\alpha < 1/\sqrt{2}$, a consistent GADF sign convention, and the presence or reliable identification of an angular boundary. Stating these requirements is essential for the paper's scope: the decoder is a controlled inverse map, not a guarantee that every visually plausible GADF image corresponds to a valid physical signal. A quantisation caveat bounds the *worst case* rather than the operating point: the entry between the two extreme samples reaches $|G^{\mathrm{D}}| = \beta = \sin\big(\arccos(-\alpha) - \arccos\alpha\big) \approx 0.9998$, where arcsin has derivative ≈ 50, so one 8-bit PNG step ($\approx 2/255$ in G^{D}: the matrix spans $[-\beta, \beta]$ but is stored and decoded over the fixed $[-1, 1]$ range across the 256 levels) there maps to an angular error of order 0.4 rad. The multi-pivot decoder, however, mostly avoids that entry: it reconstructs each trajectory from central rows whose arcsin arguments are small in typical windows (on WISDM, mean $|G^{\mathrm{D}}_{pj}| = 0.22$, 99th percentile 0.76) and reads the extreme row through one scalar G^{D}_{ep} per pivot (mean ≈ 0.78). For almost all windows the steep near-boundary region is therefore off the reconstruction path; but because the central-pivot criterion only *tends* to place pivots near the angular centre, individual entries can still approach the boundary (max ≈ 0.99 on WISDM), so this is an empirical rather than a guaranteed property. It nonetheless explains why the measured PNG error sits far below the worst-case bound (Section 6.5: floating-point MAE $< 10^{-6}$, 8-bit PNG MAE 0.0158 at $\alpha = 0.7$ with $n_p = 5$). The decoder is thus exact in real arithmetic and, empirically, keeps the quantisation-sensitive boundary off its typical reconstruction path; image-perturbation robustness is left to future work. When these assumptions fail, a more general nonlinear least-squares reconstruction can be used:

$$\min_{\phi} \sum_{i<j} w_{ij}\left(\sin(\phi_i - \phi_j) - G^{\mathrm{D}}_{ij}\right)^2,$$

with one angle fixed to remove the global phase ambiguity and with smoothness or locality weights w_{ij} for robustness.

4.5. Markov Transition Field Reconstruction

The MTF decoder described below is, to the best of our knowledge, not preceded by a documented explicit MTF-to-time-series decoder in the literature reviewed here. The decoder is not bijective: a Markov chain induces an

equivalence class of time series with the same transition statistics, and no algorithm can distinguish elements of that class from the MTF image alone. What follows recovers a representative of that class by clustering the MTF rows into quantile bins and fixing the bin ordering through the Fiedler vector of the recovered transition graph. This spectral ordering step is motivated by the seriation literature, where the Fiedler vector can recover a latent linear order under appropriate structural conditions [16]; in our setting, the exact-order statement should therefore be read as an ideal birth–death-chain case, not as a universal guarantee for arbitrary MTF images.

The reconstruction proceeds in four steps:

- **Step 1: Bin recovery.** Cluster the T rows of the MTF matrix $M \in \mathbb{R}^{T\times T}$ into Q groups by farthest-point-initialised k-means on the row-normalised image. Under a clean MTF every sample in the same quantile bin has the same row pattern, so this recovers the bin label sequence $\{q_1, \ldots, q_T\}$ up to a bin permutation.
- **Step 2: Bin ordering via the Fiedler vector.** Estimate the bin transition matrix W as the average of MTF entries inside every (p, q) block, symmetrise it as $S = \frac{1}{2}(W + W^\top)$, and order the bins by the Fiedler vector of the graph Laplacian $L = D - S$ (the eigenvector corresponding to the second-smallest eigenvalue). For an ideal birth–death Markov chain on Q ordered states this recovers the linear amplitude order; for general chains it is a spectral seriation heuristic whose quality depends on how strongly the recovered transition graph reflects an underlying one-dimensional amplitude order.
- **Step 3: Amplitude assignment.** For every time index i belonging to sorted bin G_k, assign the midpoint of the corresponding quantile interval as the reconstructed normalised value:

$$\hat{x}_i = \frac{k - \frac{1}{2}}{Q}, \qquad k = 1, \ldots, Q.$$

- **Step 4: Normalised output.** Steps 1–3 already return the reconstruction in the normalised domain (bin-midpoint values in $[0, 1]$); recovery of the physical scale, when required, follows the common calibration $\mathcal{N}_m^{-1}$ of Section 4.1.

The procedure does not verify the global sign of the series, because the MTF representation emphasises transition structure rather than signed amplitude order; a single sign-bit per axis (not stored here) would resolve the remaining mirror ambiguity. We use $Q = 4$ bins throughout the reported experiments. The choice is defended both by a noise-of-estimator argument – with $T = 128$ samples and Q bins there are $\sim (T-1)/Q^2$ transitions per cell of the empirical row-stochastic matrix W, so $Q = 4$ admits ~ 8 transitions per cell while $Q = 8$ drops to ~ 2 and many cells of W become degenerate at 0 or 1 – and empirically by the bin-count ablation in Section 6.7, where $Q = 4$ is the value that simultaneously minimises the pointwise MAE, maximises the pointwise Pearson correlation, and maximises the amplitude-sensitive KNN accuracy (CNN nearly unchanged between $Q = 2$ and $Q = 4$). The remaining classifiers (SVM, Random Forest, LSTM, GRU) vary by at most 0.012 accuracy across the entire sweep, so the recommendation of $Q = 4$ is dominated by the amplitude-sensitive subset rather than overridden by the others.

5. Experimental Protocol

The experimental protocol is designed to answer three separate questions: whether the signal can be reconstructed point-by-point, whether its global trend is preserved, and whether the reconstructed window still supports activity recognition. These questions are intentionally separated because a representation may be poor for sample-level reconstruction while still retaining enough coarse structure for classification. This section fixes the dataset, segmentation, qualitative checks, metrics, and classifier protocol before the results are reported.

5.1. Dataset and Segmentation

The dataset used is the *WISDM Smartphone and Smartwatch Activity and Biometrics Dataset* [14]. It contains data from 51 subjects, with accelerometer and gyroscope measurements collected from a smartwatch worn on the dominant hand and a mobile device kept in the pocket. The dataset includes 18 activities, each recorded for 3 minutes. We use the smartwatch accelerometer stream, which is sampled at 20 Hz (i.e., one sample every 50 ms).

For our experiments, we selected five common activities with nearly balanced sample counts. This subset keeps the classification task interpretable while avoiding a strongly imbalanced evaluation in which reconstruction effects could be masked by class-frequency effects. The chosen activities are:

We segmented the data into fixed non-overlapping windows (FNOW) of length $N = 128$, corresponding to 6.4 seconds at 20 Hz. The window length $N = 128$ (6.4 s) is a common choice for convolutional and recurrent backbones in human-activity recognition, and as a power of two it keeps the GASF, GADF, and MTF images at 128×128 pixels.

Table 3: Distribution of activities in the dataset.

Activity	Raw sensor samples	Percentage
Standing	216,529	20.6%
Sitting	213,018	20.2%
Walking	210,495	20.0%
Stairs	207,312	19.7%
Jogging	205,787	19.5%

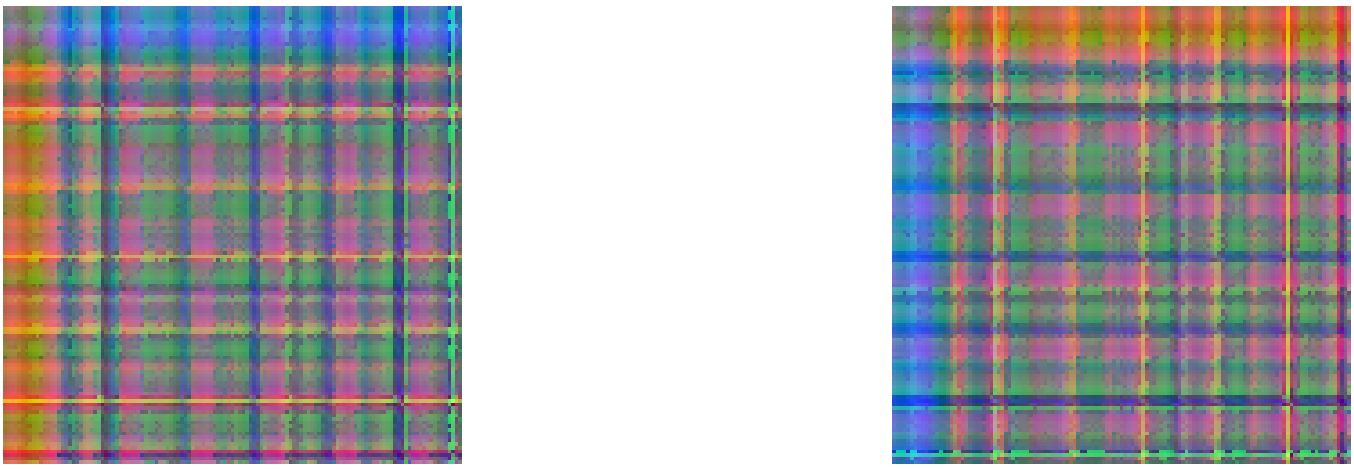
For the modified RP formulation, adjacent states are still written as $s_j = (x_j, x_{j+1})$, so a length-N window yields an $(N-1) \times (N-1) = 127 \times 127$ image; the reconstruction pipeline then recovers the N samples directly from the overlapping adjacent states (Section 4.2) so that the final signal matches the 128-sample window length used by the other encodings. With $N_{\text{bins}} = 4$ the forward MTF binning is by amplitude value-quantiles, so each of the four bins contains *approximately* $N/Q = 32$ time indices — exactly 32 only in the absence of values tied at a quantile edge.

By decomposing the X, Y, and Z dimensions into individual time series of size 128, we obtain 8 172 windows per channel, resulting in a total of $8\,172 \times 3 = 24\,516$ one-dimensional reconstructions for the reconstruction metrics. For classification, however, the experimental unit is the full triaxial window: the three reconstructed axes are grouped back into a single sample, yielding 8 172 labeled windows. The stratified window-level protocol assigns these to three folds of ∼2 724 test windows each, so the pooled paired-sample size used by the inferential tests in Section 5.3.1 is $N = 8\,172$.

Windowing and grouping details. Windowing is performed *independently within each (subject, activity) segment*: inside one contiguous recording of a given activity by a given subject we take non-overlapping blocks of $N = 128$ samples (stride 128) and discard the trailing remainder shorter than 128. No window therefore crosses a subject or an activity boundary, and every window carries an unambiguous single label and subject id. Dropping the per-segment remainders is what reduces the naive global-slide count to the 8 172 windows used here, and it removes the boundary-straddling windows that a global slide would otherwise mislabel. We evaluate two complementary 3-fold protocols. The stratified window-level split uses a grouping key that is effectively unique per window, so it reduces to a class-stratified partition at the window level: although no single window ever appears in both train and test, different — and often temporally adjacent — windows from the same subject and activity are distributed across the two sets. It therefore measures robustness to reconstruction-induced distribution shift *within* the observed population rather than trial- or subject-level generalisation, and its accuracies should be read as an optimistic within-distribution bound. The grouped subject-out split instead uses the subject id as the grouping key, so every window of a given subject falls entirely in one fold and no test subject is seen during training; this is the conservative protocol for generalisation to unseen individuals.

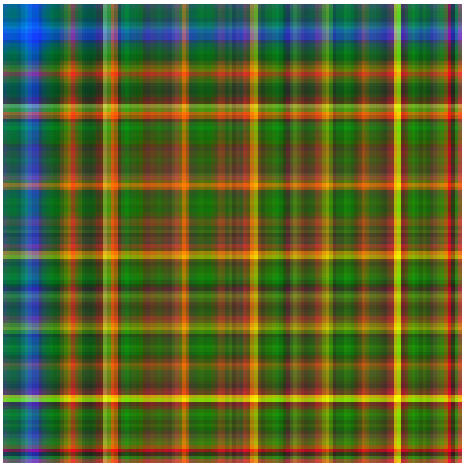
5.2. Encoding Sensitivity to Signal Inversion

Before evaluating reconstructions, we first test whether each encoding visually distinguishes a time series from its sign-inverted counterpart. This check is important because an encoding that maps both signals to the same image cannot by itself resolve orientation during reconstruction. The same WISDM window (subject 1600, first walking window, $N = 128$) is used for every encoder so the only difference across panels is the encoder itself; the inverted variant negates all three accelerometer channels.

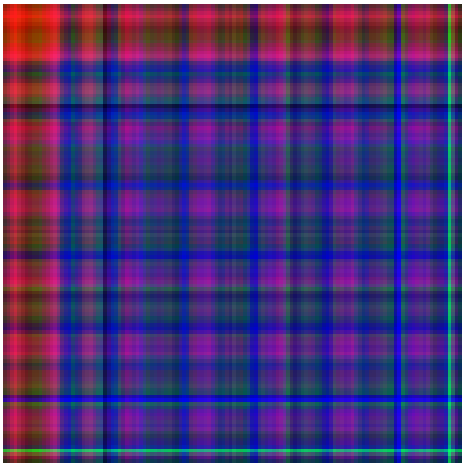


(a) Original (b) Inverted

Figure 5: Recurrence Plot (RP) images: original vs. inverted.

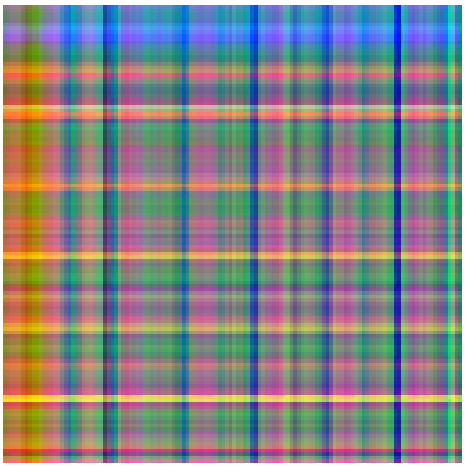


(a) Original

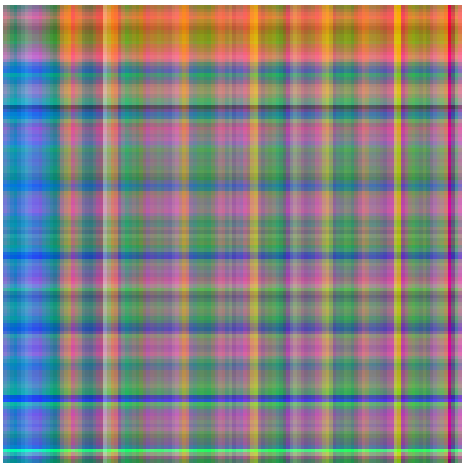


(b) Inverted

Figure 6: Gramian Angular Summation Field (GASF) images: original vs. inverted.

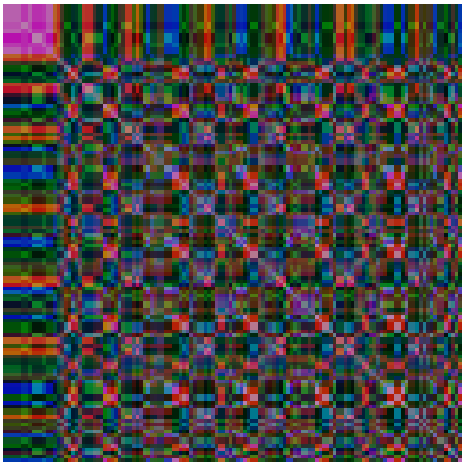


(a) Original

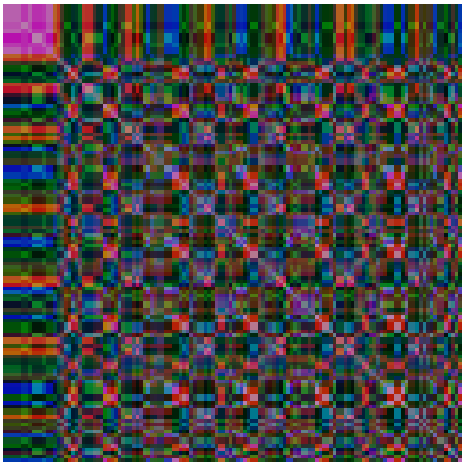


(b) Inverted

Figure 7: Gramian Angular Difference Field (GADF) images: original vs. inverted.



(a) Original



(b) Inverted

Figure 8: Markov Transition Field (MTF) images: original vs. inverted.

As illustrated in Figures 5 (RP) and 6 (GASF), the original and inverted time series produce distinctly different images. The same holds for the Gramian Angular Difference Field (Figure 7): negating the input flips the sign of $\sin(\phi_i - \phi_j)$, which after normalisation appears as a contrast inversion of the GADF image. Conversely, as shown in Figure 8, the Markov Transition Field (MTF) yields identical representations for both the original and inverted series, failing to capture their inversion-specific differences. This invariance is *not* because the transition statistics are themselves unchanged: a global sign flip reverses the quantile ranks, $q \mapsto Q{+}1{-}q$, which reverses both the bin

ordering and the rows and columns of the transition matrix, $W_{p,q} \mapsto W_{Q+1-p,\,Q+1-q}$. The two reversals cancel at the pixel level — $M_{a,b} = W_{q_a,q_b}$ maps to $W_{Q+1-(Q+1-q_a),\,Q+1-(Q+1-q_b)} = W_{q_a,q_b}$ — so the image is left unchanged. The MTF therefore discards the global sign by construction, consistent with the missing sign bit noted in Section 4.5.

5.3. Evaluation Design

We evaluated the four reconstruction pipelines with a three-tier design. The first tier measures sample-level fidelity, the second tier measures trend-level similarity, and the third tier measures downstream discriminative utility. This structure is important because the three questions are not equivalent: a method can have large pointwise errors but still preserve enough temporal structure for a classifier, or it can reconstruct amplitudes accurately while losing information that matters for a specific task. Prior work on regression error metrics and time-series similarity supports using complementary measures when each metric is tied to a specific interpretation [15, 17].

In the first tier, MAE, MRE, MSE, and Pearson correlation are used to compare the original and reconstructed samples directly. MAE gives an interpretable average absolute deviation, MSE penalizes large local failures, MRE normalizes error by signal magnitude, and Pearson correlation measures whether the temporal pattern is linearly aligned even when amplitudes differ. No single metric is sufficient, so these values are interpreted together rather than collapsed into one score.

In the second tier, each min–max normalized window is summarized by a global cubic polynomial. The coefficient vector is (a_3, a_2, a_1, a_0) in the standard power basis, not a piecewise cubic-spline representation. This choice intentionally discards local sample-level detail and tests whether the reconstructed signal preserves broad trend shape. Euclidean and Manhattan distances measure coefficient magnitude differences, cosine similarity measures directional agreement, and Mahalanobis distance accounts for the covariance structure of the original coefficient vectors [18].

In the third tier, we measure original-trained classifier robustness to reconstructed inputs. Each classifier is trained once on the original triaxial windows of the training split and then evaluated, without retraining, on the original and reconstructed versions of the same held-out windows. Classification is included because reconstructed signals should retain task-relevant information, but this experiment measures distribution shift relative to a raw-signal classifier; it is not a direct augmentation experiment. Table 4 separates the scenario reported in this paper from the additional train/test regimes that would be needed to claim augmentation utility. The per-window encoder metadata reused by the decoders (per-axis min–max, the MTF quantile interval, the sign conventions, and α) is part of this controlled inverse problem and must not be interpreted as an augmentation setting, where such metadata would be unavailable.

We state the evaluation domain explicitly, because it differs by tier. The pointwise-error tier (tier one), the visual overlays, and the classifier tier (tier three) are all computed *after* the optional physical-scale map $\mathcal{N}_m^{-1}$ has been applied: reconstructions are expressed in the original accelerometer units, the classifiers are trained and evaluated on those units, and the reported MAE/MRE/MSE are therefore directly comparable to the native amplitude range of the signal. The polynomial-trend tier (tier two) is the sole exception: its cubic coefficients are fitted on min–max-normalised windows so that the trend comparison is scale-invariant, hence its distances live in the normalised domain by construction. This split is deliberate — pointwise magnitude only means something in physical units, whereas trend shape is most fairly compared after removing per-window scale.

Table 4: Classification train/test regimes and the claim each one supports. The reported classifier tables use scenario B; scenarios C and D are listed for completeness and are *not* evaluated in this work, so no data-augmentation benefit is claimed.

Scenario	Train	Test	Main question measured
A	Original	Original	Raw-signal classifier baseline
B	Original	Reconstructed	Robustness to reconstruction-induced distribution shift
C	Reconstructed	Reconstructed	Utility inside the reconstructed domain
D	Original + Reconstructed	Original	Data-augmentation benefit for raw-signal classification

5.3.1. Statistical Significance Protocol

Every comparison reported in this paper is paired. For each stratified window-level fold, the four reconstructions and the original signal are evaluated on the same ∼2 724 test windows, and each classifier is fit once per fold and reused across the five sources within that fold. Per-fold paired correctness vectors are then concatenated across folds 0, 1 and 2, yielding a pooled paired sample of $N = 8\,172$ windows that drives all the inferential tests. Pooling is admissible here as an approximate paired comparison because each window is a single paired observation – it never appears both in training and in test within the same fold, and the test set of a fold never contains a window from any other fold's test set. One caveat remains: because each fold is scored by its own separately-trained model, the pooled 2×2 table aggregates discordant pairs produced under three different decision rules, so the homogeneity assumption behind a

single McNemar table holds only approximately and the resulting p-values should be read as mildly anti-conservative. A stratified Cochran–Mantel–Haenszel or per-fold McNemar would be the more conservative alternative; in practice every reported comparison is either a decisive separation (all RP/GASF/GADF-vs-MTF and Original-vs-MTF tests reach p_{Holm} well below 10^{-9}) or a clear non-rejected tie (all Original-vs-RP, Original-vs-GASF and Original-vs-GADF tests return $p_{\text{Holm}} = 1.0$), so this caveat changes none of the reported conclusions: no comparison sits near the significance threshold. A second concern is that the pooled window sample is not fully independent — many test windows share a subject, so window-level p-values could in principle be inflated by intra-subject pseudoreplication. We therefore repeat the pointwise comparison at the subject level, collapsing each subject to a single mean-MAE value so that the $N = 51$ subjects, rather than the 8 172 windows, are the independent units (reported in Section 7.3).

For *accuracy reporting*, every accuracy cell in the per-classifier tables (Tables 22, 23, 24, 25, 26, and 27) is the mean of the three fold accuracies and the dispersion is reported as the standard deviation across folds. Fold-level dispersion is a more honest measure of generalisation than a within-fold bootstrap, which only captures sampling noise inside a single split.

For *paired classifier comparisons* we use the continuity-corrected McNemar test on the pooled discordant pairs [20] – the standard test when two classifiers are evaluated on the same instances – and we distinguish two test families:

- **$\mathbf{F_{M1}}$: Original vs. reconstruction.** For every one of the 4 reconstructions (RP, MTF, GASF, GADF) and every one of the 6 classifiers we test whether the paired correctness vector differs from the original-signal baseline. This produces $4 \times 6 = 24$ paired tests. These McNemar tests are degradation/difference tests, not formal equivalence tests; absence of significance is therefore reported as "no detected degradation" rather than proof of equivalence.
- **$\mathbf{F_{M2}}$: encoder vs. encoder at fixed classifier.** For every pair of reconstructions and every classifier we ask which encoder yields the higher accuracy under that classifier. There are $\binom{4}{2} = 6$ encoder pairs and 6 classifiers, hence $6 \times 6 = 36$ paired tests.

For *paired pointwise-error comparisons* between encoders we use the Wilcoxon signed-rank test on per-window MAE [21]. This test compares two reconstructions on the same windows and does not involve a classifier, so the family contains only $\binom{4}{2} = 6$ tests ($\mathrm{F_W}$). Alongside each Wilcoxon p-value we report a non-parametric effect size $r = Z/\sqrt{N}$, where $|r| \geq 0.5$ is conventionally read as a large effect [22]. The specific GASF-vs-GADF fidelity gap is assessed through this same paired per-window MAE test and, to guard against intra-subject pseudoreplication, through a subject-clustered Wilcoxon that collapses each subject to a single mean-MAE value ($N = 51$ independent units).

Because the three families $\mathrm{F_{M1}}$, $\mathrm{F_{M2}}$ and $\mathrm{F_W}$ contain 24, 36 and 6 simultaneous tests respectively, raw p-values within each family are corrected with the Holm–Bonferroni step-down procedure [23] at a family-wise error rate $\alpha = 0.05$. Throughout the paper we report the Holm-adjusted value p_{Holm} and call a comparison significant only when $p_{\text{Holm}} < \alpha$ inside its own family; the unadjusted p-values are not used to declare significance. This controls the family-wise error rate *within* each family at 0.05, not a single experiment-wide rate across all three families: treating the families as independent, the across-family error is approximately $1 - (1 - 0.05)^3 \approx 0.14$, so a comparison that is only marginally significant inside its family should not be read as protected at the 0.05 level once the three families are considered jointly. We keep per-family control deliberately, because the families answer distinct questions (degradation relative to the original, encoder-vs-encoder ranking, and pointwise ranking) and are reported and interpreted separately.

5.3.2. Pointwise Error Metrics

To evaluate the reconstruction methods, we use the following error metrics:

- **Mean Absolute Error (MAE):** This metric represents the average of the absolute differences between the original and reconstructed values. It provides a direct measure of the magnitude of the errors without considering their direction.
- **Mean Relative Error (MRE):** Calculated as

$$\text{MRE} = \frac{1}{n} \sum_{i=1}^{n} \frac{|x_i - \hat{x}_i|}{|x_i| + \varepsilon},$$

  where $\varepsilon = 10^{-8}$ prevents division by zero when accelerometer values are zero or close to zero. This metric normalizes the absolute error by the magnitude of the original signal.
- **Mean Squared Error (MSE):** By squaring each error before averaging, the MSE gives a higher penalty to larger errors, thereby emphasizing large deviations in the reconstruction.
- **Pearson Correlation Coefficient:** This statistical measure indicates the degree of linear correlation between the original and reconstructed signals. A coefficient close to 1 indicates that the reconstructed signal follows the original temporal pattern.

These metrics are computed for each accelerometer axis (Dim 0, Dim 1, and Dim 2) and then aggregated as a global mean, allowing axis-level errors and overall reconstruction quality to be reported separately for RP, MTF, GASF, and GADF.

5.3.3. Polynomial Trend Metrics

To evaluate the performance of the polynomial representation reconstruction, the following normalized metrics were computed on the cubic coefficient vectors. Each coefficient dimension was standardized using the mean and standard deviation estimated from the original windows before distances were calculated. For Mahalanobis distance, the covariance matrix was estimated from the standardized coefficient vectors of the original windows and regularized as $\Sigma_\lambda = \Sigma + 10^{-6} I$ before inversion.

- **Mean Euclidean Distance:** This metric calculates the average Euclidean distance between the standardized cubic coefficient vectors of the original and reconstructed signals. A lower value indicates a more accurate reconstruction.
- **Mean Manhattan Distance:** This metric represents the mean absolute difference (L1 norm) between the standardized cubic coefficient vectors, and it is sensitive to the overall magnitude of the differences.
- **Mean Cosine Similarity:** This metric evaluates the cosine of the angle between the standardized coefficient vectors, yielding a value close to 1 when the reconstructed polynomial follows the same direction as the original, regardless of magnitude differences.
- **Mean Mahalanobis Distance:** This metric measures the distance between the standardized coefficient vectors using Σ_λ^{-1}, taking into account the variance and correlation structure estimated from the original coefficient vectors. It provides a multidimensional assessment of the similarity between the original and reconstructed polynomials.

The quantitative results for these metrics are presented in Table 8 later in the document.

5.3.4. Classifier Evaluation

To test whether reconstructed windows retain task-relevant information, we evaluated them with several classifiers. These classifiers do not prove pointwise fidelity; instead, they indicate whether class-discriminative temporal structure remains available after reconstruction. The selection of classifiers was inspired by Ignatov et al. [19], who analyzed multiple classifiers on accelerometer-based activity recognition.

Specifically, we used a three-fold protocol with $n = 3$ partitions. Each fold holds out one partition for testing and uses the remaining two partitions for training, producing a 66.6%/33.3% train-test split per fold. The 8,172 instances correspond to triaxial windows, not to separate one-dimensional axes; therefore, each fold contains approximately 2,724 test windows. Reported metrics represent the average performance obtained across the three folds when comparing the original time series against their reconstructed counterparts.

A detailed overview of each classifier and its intended role is presented below:

- **Gated Recurrent Unit (GRU):** A recurrent neural network variant designed to effectively capture temporal dependencies in sequential data. Its gating mechanisms regulate the information flow, enabling recognition of complex patterns over time.
- **K-Nearest Neighbors (KNN):** An instance-based classifier that assigns labels based on the majority class of the closest k neighbors. It acts as a baseline to determine how well separated the reconstructed classes are within the feature space.
- **Convolutional Neural Network (CNN):** A deep neural network architecture optimized for extracting local temporal patterns and spatial relationships through convolutional filters. CNN effectively identifies discriminative motifs critical for activity classification.
- **Random Forest (RF):** An ensemble classifier comprising multiple decision trees that independently vote to determine the final class prediction. RF robustly captures nonlinear relationships and interactions among features.
- **Support Vector Machine (SVM):** This classifier finds separating hyperplanes in high-dimensional feature spaces and tests whether reconstructed windows remain separable under a margin-based model.
- **Long Short-Term Memory (LSTM):** A recurrent neural network designed to model long-term dependencies in sequential data. Its gating mechanisms test whether longer temporal correlations remain available after reconstruction.

The exact configuration of each classifier is summarised in Table 5. To make the neural-network results reproducible, the global random seed (Python, NumPy and TensorFlow) is fixed to 95; K-Nearest Neighbors and Support Vector Machine are deterministic given the training data, and Random Forest uses the same value as its `random_state`.

Table 5: Classifier configurations. Each model is trained once per fold on the original (un-reconstructed) windows and reused across all four encodings. Input is $(128, 3)$ for the neural models and the flattened 384-vector for the scikit-learn models; all neural models use the sparse categorical cross-entropy loss.

Classifier	Configuration	Seed
KNN	$k = 5$, uniform weights, Minkowski distance ($p = 2$)	—
SVM	RBF kernel, $C = 1.0$, $\gamma =$ scale	—
Random Forest	100 trees, Gini split, unlimited depth, `max_features`$= \sqrt{d}$	95
LSTM	LSTM(64) $\to$ Dropout(0.2) $\to$ LSTM(32) $\to$ Dropout(0.2) $\to$ Dense-softmax; Adam (10^{-3}), 20 epochs, batch 64	95
GRU	GRU(64) $\to$ Dropout(0.2) $\to$ GRU(32) $\to$ Dropout(0.2) $\to$ Dense-softmax; Adam (10^{-3}), 20 epochs, batch 64	95
CNN	Conv1D(196, kernel 16) + MaxPool(4) concatenated with per-channel statistical features (mean, variance, $\sum \lvert \cdot \rvert$, 10-bin histogram); Dense(1024) + Dropout(0.05) + Dense-softmax; $\ell_2 = 10^{-4}$; Adam (5×10^{-4}), 20 epochs, batch 64	95

6. Results

The results are presented from the most direct evidence to the most indirect: visual reconstruction, pointwise error tables, polynomial trend similarity, and downstream classification. This order makes clear when a method reconstructs the signal itself and when it merely preserves information useful for a classifier.

6.1. Qualitative Reconstruction Analysis

6.1.1. Artificial Time Series

For our first experiment, we construct a synthetic time series $\mathbf{w} \in \mathbb{R}^{N\times 3}$ of length $N = 128$ by sampling an underlying parameter t uniformly over $[0, \pi]$. Concretely, define

$$t_i = \frac{i}{N-1}\,\pi, \qquad i = 0, 1, \ldots, N-1,$$

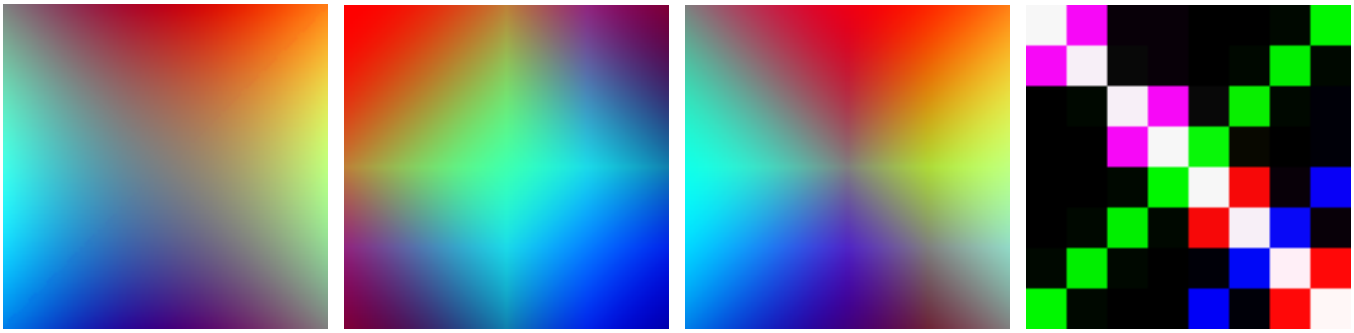
and let each vector $\mathbf{w}_i = [w_{i,0},\, w_{i,1},\, w_{i,2}]^\top$ be given by

$$w_{i,0} = \cos(t_i), \qquad w_{i,1} = \sin(t_i), \qquad w_{i,2} = \sin(t_i) \;-\; \cos(t_i).$$

Thus the full data matrix is

$$\mathbf{w} = \begin{bmatrix} \cos(t_0) & \sin(t_0) & \sin(t_0) - \cos(t_0) \\ \cos(t_1) & \sin(t_1) & \sin(t_1) - \cos(t_1) \\ \vdots & \vdots & \vdots \\ \cos(t_{N-1}) & \sin(t_{N-1}) & \sin(t_{N-1}) - \cos(t_{N-1}) \end{bmatrix}.$$

This three-dimensional signal, composed of a pure cosine channel, a pure sine channel, and their difference, serves as a controlled sanity check for the RP, GASF, GADF, and MTF reconstructions. It is not intended to represent the full WISDM distribution; instead, it provides a visually interpretable case where phase, amplitude, and orientation errors are easy to detect. To keep the evidence separated, every figure in this subsection refers only to the artificial control signal. Figure 9 shows the image encodings of the three artificial channels under each of the four techniques (RP is 127×127 because of the $N - 1$ pair embedding; the others are 128×128).



(a) RP (b) GASF (c) GADF (d) MTF

Figure 9: Image encodings of the same artificial three-channel signal (cosine, sine, sine−cosine) under each technique.

After generating the encoded images, we decode them and compare the reconstructed series graphically.

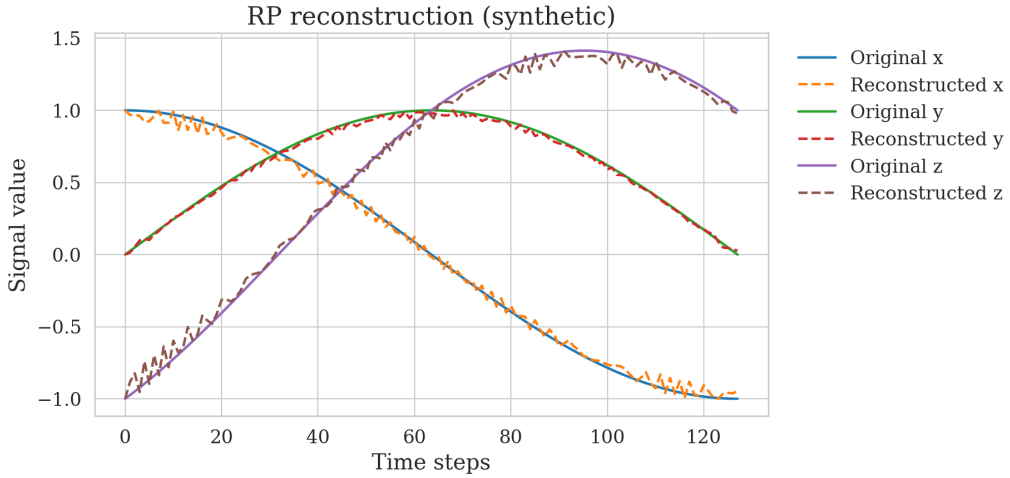


Figure 10: RP reconstruction on the artificial triaxial time series.

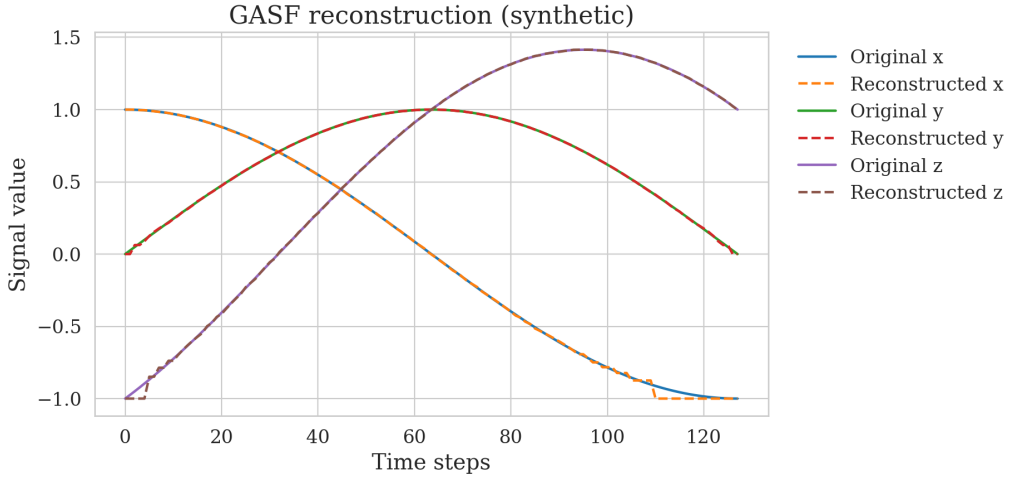


Figure 11: GASF reconstruction on the artificial triaxial time series.

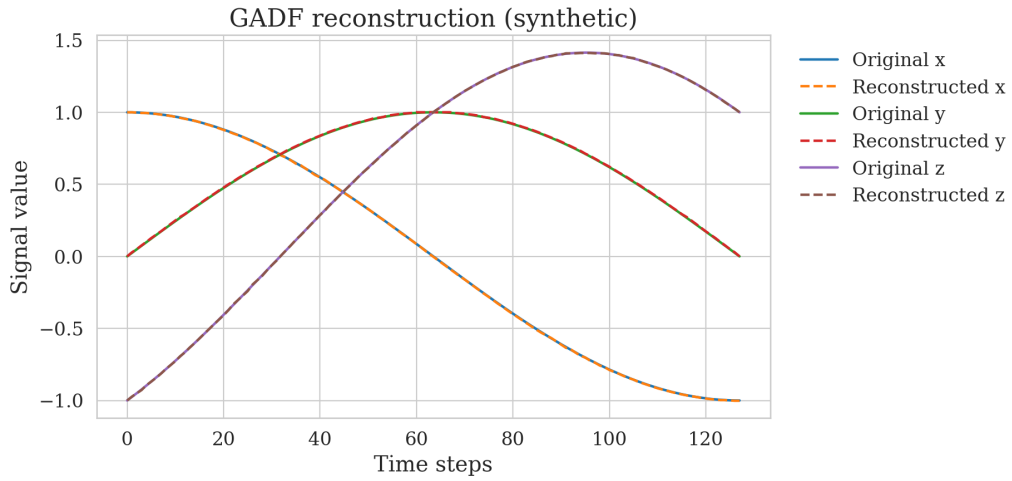


Figure 12: GADF reconstruction on the artificial triaxial time series.

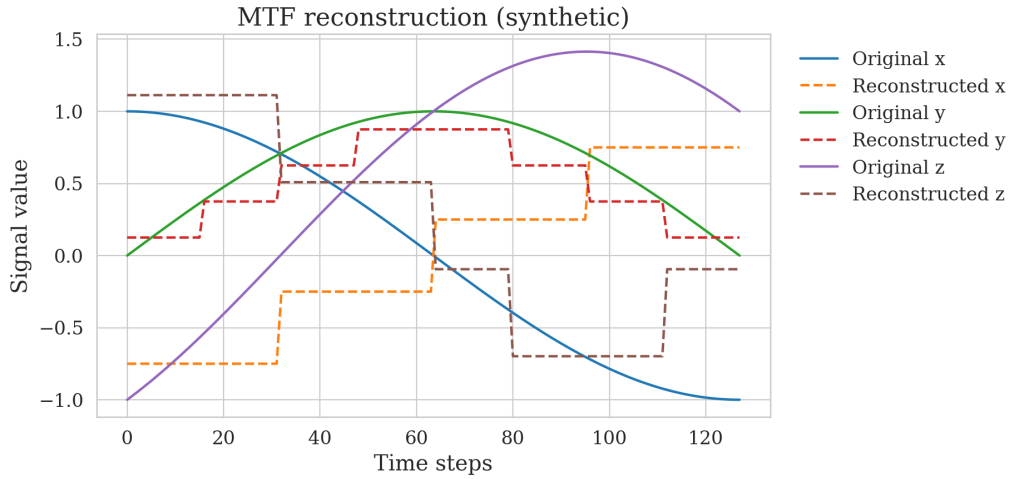
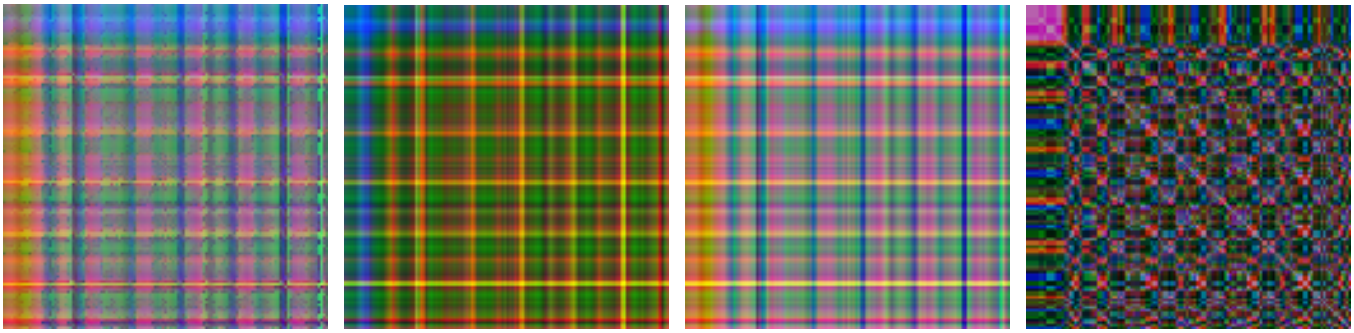


Figure 13: MTF reconstruction on the artificial triaxial time series.

After encoding the artificial time series (Figure 9), we reconstruct each representation and overlay it with the original signal. In this controlled example, the RP comparison (Figure 10) overlays the original almost exactly, recovering both shape and amplitude. The GASF comparison (Figure 11) is visually closest to the original in all three channels under the tested normalization. The GADF comparison (Figure 12) is likewise near-perfect: the multi-pivot, boundary-anchored reconstruction overlays the cosine, sine and sin−cos channels almost exactly, confirming that the restricted-branch inverse recovers the synthetic signal up to image quantisation. By contrast, the MTF comparison (Figure 13) reveals two limitations that also matter in real data: possible orientation inversion and discretization errors caused by the four transition bins.

6.1.2. WISDM Sample `1600×0`

To assess encoding–decoding fidelity on real sensor data, we selected sample `1600×0` from fold 0 of the WISDM training set (class 0, Walking, $N = 128$ samples). This example is used for qualitative interpretation only; aggregate claims are based on the quantitative tables that follow. Figure 14 shows the encoded images of *the same time-series window* under each of the four encoding techniques, with the three accelerometer axes (x, y, z) mapped to the red/green/blue channels of the image. The RP image is 127×127 because the modified definition $s_j = (x_j, x_{j+1})$ embeds the window into $N - 1$ adjacent states; GASF, GADF, and MTF are 128×128.



(a) Recurrence Plot (RP) (b) GASF (c) GADF (d) Markov Transition Field (MTF)

Figure 14: The four image encodings of the same WISDM window `1600×0` (fold-0 train, Walking). Per-axis (x,y,z) intensities are stacked as the (R,G,B) channels of each image.

After encoding, each image is decoded back into a three-axis time series and overlaid on the original signal. Figures 15–18 present the global comparison across all four encoders (RP, GASF, GADF, MTF).

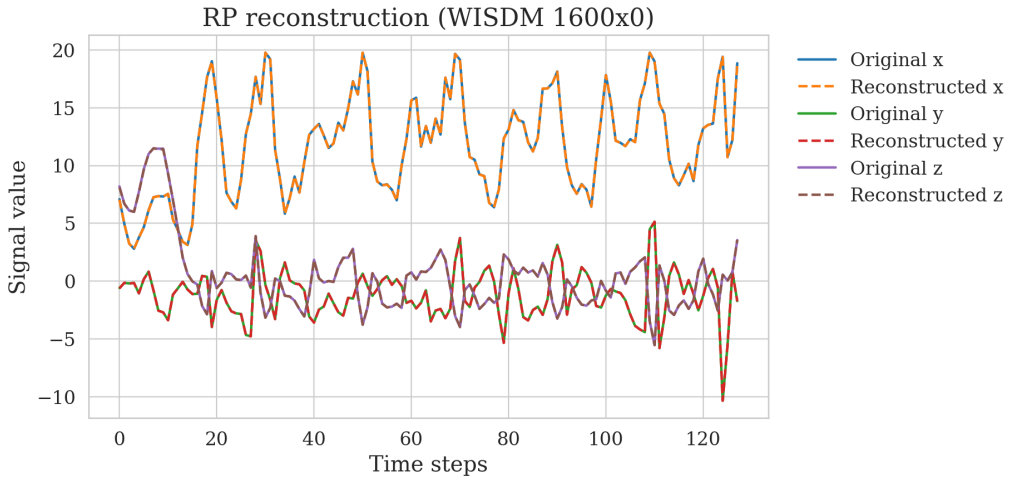


Figure 15: Original vs. RP reconstruction for sample `1600×0`.

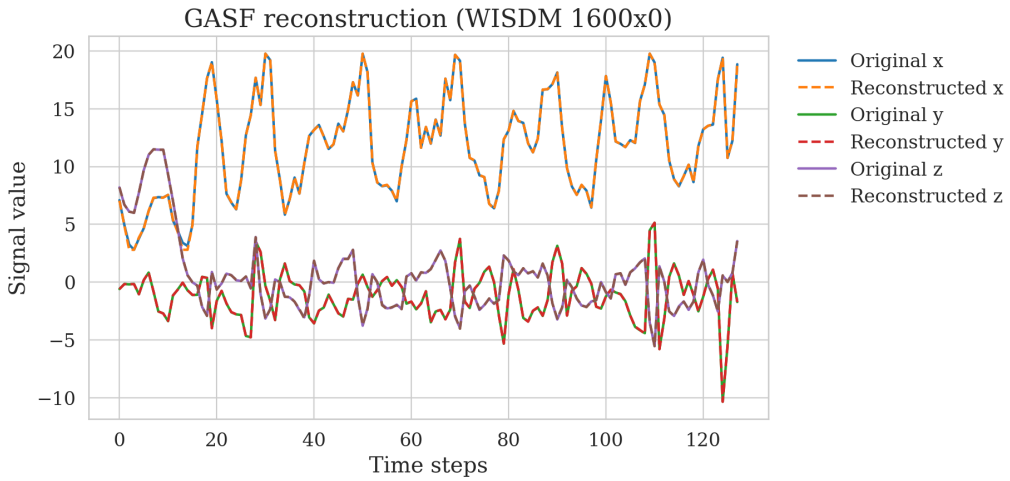


Figure 16: Original vs. GASF reconstruction for sample `1600×0`.

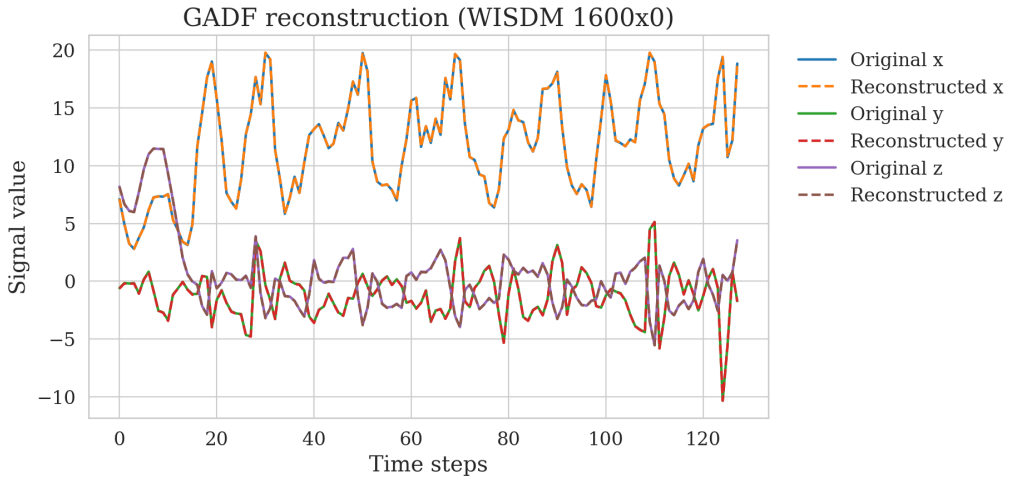


Figure 17: Original vs. GADF reconstruction for sample `1600×0`.

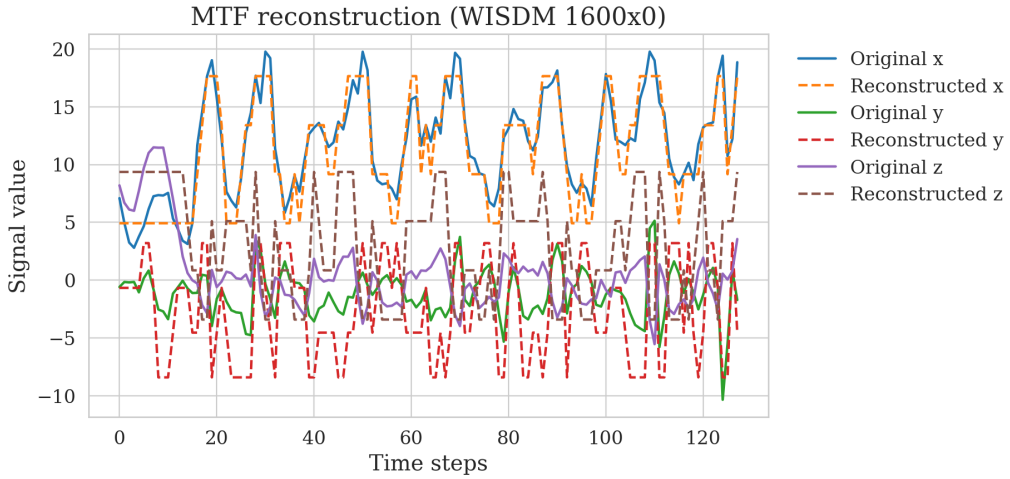


Figure 18: Original vs. MTF reconstruction for sample `1600×0`.

Across the four evaluated methods, GASF (Fig. 16) produces the closest visual reconstruction under the tested protocol, with decoded signals nearly indistinguishable from the original. GADF (Fig. 17) is visually on par with GASF: its multi-pivot, boundary-anchored decoder recovers both the global shape and the short-range variation, consistent with its near-original pointwise error and the absence of any detected classifier degradation in Section 7.5. RP (Fig. 15) is likewise near-indistinguishable from the original: with the corrected inverse normalisation, the 2-D classical-MDS embedding, the overlap reconstruction, and the signed-orientation rule recover both shape and amplitude, leaving only quantisation-level error. MTF (Fig. 18), despite using only four quantised transition levels, captures the general shape but shows larger deviations in segments where neighboring transition states are difficult to order.

For clarity, Figure 19–21 split the MTF comparison per axis:

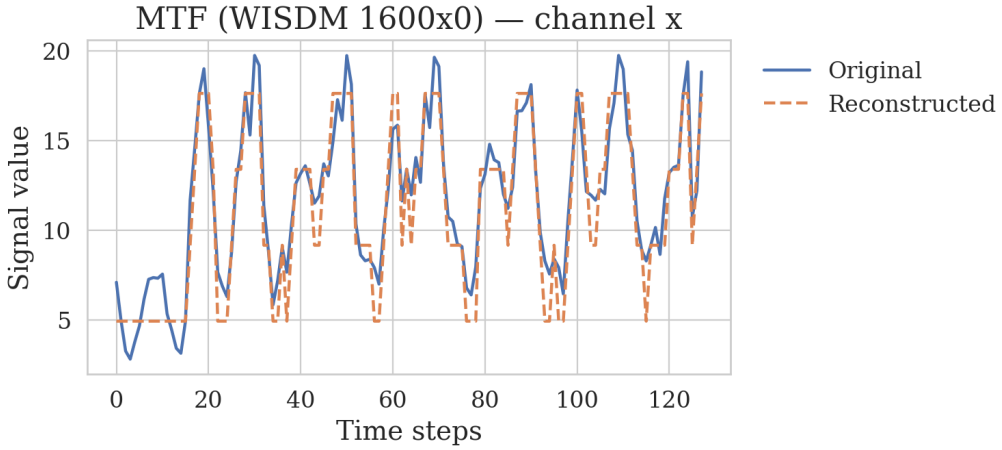


Figure 19: Original vs. MTF reconstruction for sample `1600×0`, axis x.

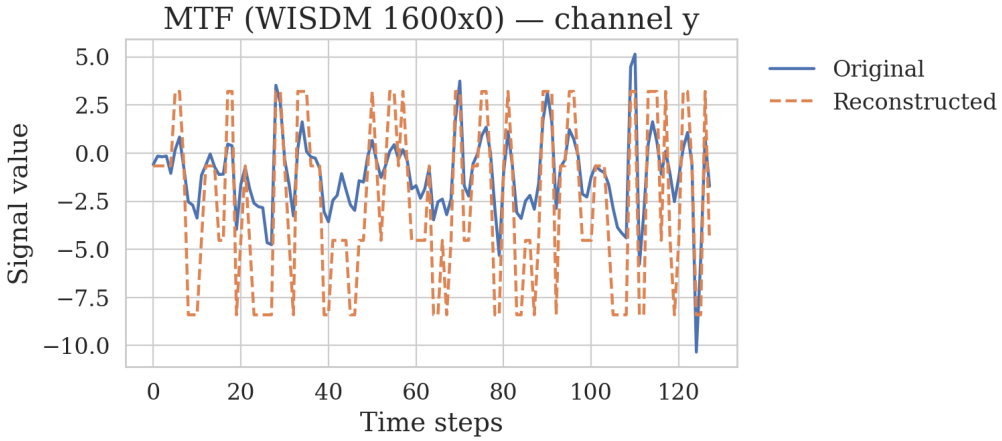


Figure 20: Original vs. MTF reconstruction for sample `1600×0`, axis y.

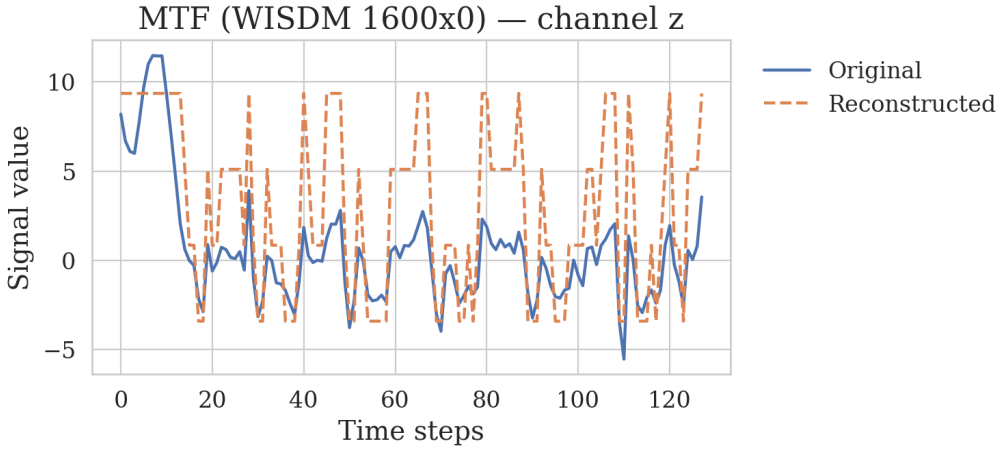


Figure 21: Original vs. MTF reconstruction for sample `1600×0`, axis z.

6.2. Quantitative Reconstruction Results

6.2.1. Pointwise Error Metrics

Unless otherwise specified, all error calculations were performed on the raw (unnormalized) time series in the dataset's native accelerometer units (m/s^2), so as to quantify the absolute magnitude of reconstruction errors in the context of the data's native amplitude range. Because the decoders are deterministic and untrained, a window reconstructs identically regardless of the fold or protocol it lands in: the three test folds of each 3-fold protocol simply repartition

the same 8172 windows, so their union is the full dataset with every window scored exactly once. We therefore report the error for *every fold of both protocols* in Table 6 — the stratified window-level and the grouped subject-out 3-fold splits alike — alongside the pooled full-dataset aggregate in Table 7, which is numerically identical for the two protocols and coincides with the population used by the paired significance tests of Section 7.3. This protocol-independence of the *pooled* figure is exact by construction, not a coincidence: the two protocols assign the very same 8172 windows to folds differently, but reconstruction involves no training, so a window's decoded signal — and hence its error — depends only on the window, never on the fold or protocol it is placed in. The window-level and subject-out splits therefore differ only in how the windows are *grouped*, which changes the per-fold column (each subject-out fold holds out a disjoint set of 16–18 subjects, each window-level fold contains all 51) but leaves the pooled union unchanged. This is unlike the downstream classification of Section 7.5, where the classifier *is* trained per fold and the two protocols genuinely diverge (holding out whole subjects is the harder task). The per-fold table shows that reconstruction quality is stable across folds: the window-level folds are near-identical stratified partitions, and even the grouped subject-out folds — where entirely different subjects are held out — move only slightly. Table 7 additionally gives, per encoder and metric, the across-window median[IQR] and a subject-cluster bootstrap 95% confidence interval (the 51 subjects, not the windows, resampled); the per-axis breakdown is in Appendix A.

Table 6: Per-fold reconstruction error for both 3-fold protocols (physical units, m/s^2). Each cell is the per-window global metric (averaged over the three axes) averaged over that fold's test windows. Because the decoders are deterministic and untrained, a window reconstructs identically in any fold; the two protocols only repartition the same 8172 windows, so the Pooled column is identical for both and equals the full-dataset value of Table 7. The window-level folds are near-identical stratified partitions (negligible fold-to-fold spread), whereas the subject-out folds vary more because different subjects are held out.

		Window-level 3-fold			**Subject-out 3-fold**			
Encoder	**Metric**	F0	F1	F2	F0	F1	F2	**Pooled**
RP	MAE	0.0148	0.0149	0.0149	0.0142	0.0155	0.0147	0.0148
	MRE	0.0047	0.0048	0.0047	0.0047	0.0051	0.0044	0.0047
	MSE	0.0009	0.0011	0.0011	0.0009	0.0011	0.0010	0.0010
	Pearson	0.99998	0.99998	0.99998	0.99998	0.99998	0.99998	0.99998
GASF	MAE	0.0202	0.0205	0.0207	0.0189	0.0211	0.0214	0.0205
	MRE	0.0060	0.0061	0.0061	0.0059	0.0064	0.0059	0.0061
	MSE	0.0042	0.0043	0.0044	0.0037	0.0041	0.0051	0.0043
	Pearson	0.99988	0.99988	0.99988	0.99988	0.99988	0.99988	0.99988
GADF	MAE	0.0156	0.0157	0.0159	0.0147	0.0162	0.0163	0.0157
	MRE	0.0054	0.0055	0.0055	0.0054	0.0057	0.0053	0.0055
	MSE	0.0008	0.0008	0.0008	0.0007	0.0008	0.0009	0.0008
	Pearson	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999
MTF	MAE	3.98	3.94	3.91	3.71	4.04	4.07	3.94
	MRE	1.22	1.22	1.21	1.23	1.26	1.16	1.22
	MSE	56.42	55.25	54.62	48.13	52.54	65.43	55.43
	Pearson	0.00289	0.02089	0.01689	0.00977	0.00116	0.02982	0.01356

Table 7: Pooled pointwise reconstruction errors over all $N = 8172$ windows — the full dataset (the three test folds of either protocol pooled; identical, since both repartition the same windows; physical units, m/s^2). This aggregates the per-fold values of Table 6. Because the decoders are deterministic and untrained, the pooled set is the full dataset with each window scored once. Each metric is computed per window (averaged over the three accelerometer axes); columns report the across-window mean±std and median[IQR], and the subject-cluster bootstrap 95% CI of the grand per-subject mean (51 subjects, 10^4 resamples). Per-axis breakdowns are in Appendix A.

Encoder	Metric	Mean ± Std	Median [IQR]	Per-subject mean [95% CI]
RP	Mean Absolute Error	0.0148 ± 0.0170	0.0091 [0.0014, 0.0232]	0.0145 [0.0135, 0.0155]
	Mean Relative Error	0.0047 ± 0.0050	0.0035 [0.0006, 0.0074]	0.0046 [0.0043, 0.0049]
	Mean Squared Error	0.0010 ± 0.0046	0.0001 [0.0000, 0.0009]	0.0010 [0.0008, 0.0012]
	Pearson Correlation Coefficient	0.99998 ± 0.00005	0.99999 [0.99998, 0.99999]	0.99998 [0.99998, 0.99999]
GASF	Mean Absolute Error	0.0205 ± 0.0199	0.0158 [0.0019, 0.0330]	0.0199 [0.0188, 0.0212]
	Mean Relative Error	0.0061 ± 0.0056	0.0052 [0.0008, 0.0096]	0.0060 [0.0057, 0.0063]
	Mean Squared Error	0.0043 ± 0.0088	0.0010 [0.0000, 0.0048]	0.0040 [0.0036, 0.0046]
	Pearson Correlation Coefficient	0.99988 ± 0.00014	0.99991 [0.99985, 0.99994]	0.99988 [0.99987, 0.99988]
GADF	Mean Absolute Error	0.0157 ± 0.0167	0.0112 [0.0018, 0.0236]	0.0148 [0.0137, 0.0160]
	Mean Relative Error	0.0055 ± 0.0057	0.0042 [0.0007, 0.0083]	0.0052 [0.0048, 0.0055]
	Mean Squared Error	0.0008 ± 0.0018	0.0002 [0.0000, 0.0009]	0.0007 [0.0006, 0.0008]
	Pearson Correlation Coefficient	0.99999 ± 0.00001	0.99999 [0.99999, 0.99999]	0.99999 [0.99999, 0.99999]
MTF	Mean Absolute Error	3.94 ± 4.25	2.74 [0.40, 5.91]	3.71 [3.45, 4.00]
	Mean Relative Error	1.22 ± 1.20	1.00 [0.16, 1.89]	1.15 [1.08, 1.23]
	Mean Squared Error	55.43 ± 113.46	13.07 [0.27, 58.91]	47.30 [40.24, 56.59]
	Pearson Correlation Coefficient	0.01356 ± 0.48870	0.06274 [-0.30746, 0.31208]	0.01389 [0.00318, 0.02463]

The proposed decoder recovers the two-dimensional state geometry directly from the recovered distances and does not use the recurrence-rate parameter ρ of the earlier graph-based variant, so no recurrence-rate sensitivity analysis applies.

6.2.2. Polynomial Trend Comparison

Table 8: Normalized polynomial-trend coefficient metrics over all $N = 8172$ pooled windows.

Metric	RP	GASF	GADF	MTF
Mean Euclidean Distance	0.00081	0.00081	0.00070	0.23224
Mean Manhattan Distance	0.00083	0.00087	0.00072	0.24459
Mean Cosine Similarity	0.99991	0.99975	1.00000	0.90610
Mean Mahalanobis Distance	0.00814	0.01042	0.00721	2.85618

6.3. Downstream Classification Results

The full per-class precision, recall, F1, and support for all six classifiers — on the original validation windows and on each reconstructed signal (RP, MTF, GASF, GADF) — are reported in Appendix B; the body retains only the compact accuracy comparison (Table 16) needed to interpret the differences among reconstruction methods. Those numbers are averaged across all 3 folds of the stratified window-level protocol (mean ± std, ∼2 724 test windows per fold) and are scenario-B results in Table 4: all six classifiers share the same trained model across encodings, fitted once on the original training set, so only the prediction input changes across the GASF/GADF/RP/MTF columns.

6.4. Grouped Subject-Out Robustness

The stratified window-level 3-fold protocol used so far is a class-stratified split at the window level: because the grouping key is effectively unique per window, windows from the same subject – and indeed from the same recording – can fall on both sides of a fold. This is a common configuration for within-distribution evaluation and the right test for whether a reconstruction preserves task-relevant temporal structure within the observed population. It is not, however, a test of whether the model generalises to *unseen subjects* – the regime that matters most for wearable

deployment. We therefore replicate the full evaluation under a grouped subject-out 3-fold protocol: the 51 WISDM subjects are partitioned by StratifiedGroupKFold into three disjoint groups of 16, 18 and 17 test subjects respectively, and the test set of one fold never shares a subject with either of its training folds. We use the term subject-out rather than strict leave-one-subject-out, since each fold holds out a disjoint group of subjects rather than a single subject. Encoding, reconstruction, and the six-classifier panel are exactly the same as for the window-level protocol; only the train/test indices change. Each subject-out fold contains $\sim$2 724 test windows so the pooled paired-sample size for inferential tests is again $N = 8\,172$.

Table 9: Headline accuracies under the grouped subject-out 3-fold protocol (mean $\pm$ std across folds 0, 1, 2; $\sim$2 724 test windows per fold drawn from 16–18 held-out subjects). Δacc is the accuracy of the reconstructed signal minus the accuracy of the original signal under the same classifier, averaged over the same three folds.

Classifier	**Orig.**	**GASF**	**GADF**	**RP**	**MTF**
GRU	0.811 ± 0.026	0.811 ± 0.026 (+0.000)	0.811 ± 0.026 (+0.000)	0.811 ± 0.026 (+0.000)	0.702 ± 0.025 (−0.109)
KNN	0.504 ± 0.035	0.504 ± 0.034 (−0.000)	0.505 ± 0.034 (+0.000)	0.504 ± 0.035 (−0.000)	0.465 ± 0.039 (−0.040)
CNN	0.796 ± 0.036	0.796 ± 0.036 (−0.000)	0.796 ± 0.036 (−0.000)	0.794 ± 0.049 (−0.003)	0.686 ± 0.035 (−0.110)
Random Forest	0.786 ± 0.039	0.787 ± 0.039 (+0.001)	0.787 ± 0.039 (+0.000)	0.786 ± 0.039 (−0.001)	0.689 ± 0.026 (−0.098)
Support Vector Machine	0.738 ± 0.020	0.738 ± 0.020 (+0.000)	0.738 ± 0.020 (−0.001)	0.739 ± 0.020 (+0.000)	0.665 ± 0.026 (−0.073)
LSTM	0.744 ± 0.029	0.744 ± 0.029 (+0.000)	0.744 ± 0.028 (−0.000)	0.744 ± 0.028 (−0.000)	0.672 ± 0.035 (−0.072)

Table 10: Per-fold accuracies on the grouped subject-out 3-fold protocol. Folds split the 51 WISDM subjects into disjoint groups of 16, 18 and 17 test subjects respectively; the test set of one fold never shares a subject with either of its training folds.

	Orig.			**GASF**			**GADF**			**RP**			**MTF**		
Classifier	f0	f1	f2	f0	f1	f2	f0	f1	f2	f0	f1	f2	f0	f1	f2
GRU	0.848	0.791	0.794	0.848	0.791	0.794	0.847	0.791	0.794	0.848	0.791	0.794	0.732	0.671	0.702
KNN	0.551	0.495	0.467	0.550	0.494	0.468	0.550	0.496	0.467	0.551	0.495	0.466	0.515	0.458	0.421
CNN	0.847	0.779	0.764	0.847	0.778	0.764	0.847	0.778	0.764	0.859	0.740	0.782	0.735	0.670	0.654
Random Forest	0.842	0.753	0.764	0.841	0.754	0.766	0.842	0.754	0.765	0.840	0.751	0.765	0.726	0.666	0.674
Support Vector Machine	0.757	0.711	0.748	0.757	0.710	0.748	0.756	0.709	0.747	0.757	0.711	0.748	0.701	0.639	0.654
LSTM	0.779	0.708	0.745	0.779	0.708	0.745	0.778	0.709	0.745	0.777	0.708	0.745	0.719	0.633	0.664

Three observations follow from Tables 9 and 10, read together with the window-level-vs-subject-out summary that opens the discussion (Table 16):

1. **The encoder ranking is preserved under subject-out evaluation.** This is the load-bearing robustness check: the window-level orderings are not artefacts of allowing the model to memorise per-subject calibration. Under grouped subject-out validation the absolute accuracies drop by 0.06–0.14 as expected (the model has never seen the test subjects), but the three near-original decoders remain statistically tied with the raw signal on every classifier: the paired McNemar test fails to reject Original-vs-RP, Original-vs-GASF and Original-vs-GADF in all six classifiers ($p_{\text{Holm}} = 1.0$ throughout). MTF is again the single decoder that degrades, losing 0.04–0.11 accuracy relative to the original and being significantly outranked by RP, GASF and GADF on every classifier (p_{Holm} from 2.4×10^{-19} on KNN to 3.1×10^{-116} on GRU for RP-vs-MTF). The top tier is therefore not a within-distribution artefact, and the near-lossless behaviour of RP, GASF and GADF transfers intact to unseen subjects.
2. **GASF shows no detected degradation from the original under this paired test.** The paired McNemar test on the pooled subject-out sample fails to reject Original-vs-GASF in all six classifiers ($p_{\text{Holm}} = 1.00$ for all six); the absolute subject-out accuracy drop of GASF relative to the original is bounded by $|\Delta\text{acc}| \leq 0.001$ across the six classifiers, identical to the window-level behaviour. This replicates the near-equivalence pattern of Section 7.5, while still stopping short of a formal equivalence or information-preservation claim.
3. **No reconstruction improves on the original under subject-out; RP tracks the raw signal exactly like the Gramian fields.** On every classifier RP matches the original-signal accuracy to within noise — KNN 0.504 vs 0.504, SVM 0.739 vs 0.738, Random Forest 0.786 vs 0.786, LSTM 0.744 vs 0.744, GRU 0.811 vs 0.811, CNN 0.794 vs 0.796 — and the paired McNemar test fails to reject Original-vs-RP in all six ($p_{\text{Holm}} = 1.0$), just as it does for GASF and GADF. The corrected RP decoder therefore generalises to unseen subjects without loss, and none of the four reconstructions exceeds the original under this stricter protocol.

6.5. GADF Decoder Reliability: Exactness, Anchor, and Pivot Ensemble

We verify the two numeric choices of the GADF decoder — the amplitude restriction α and the number of averaged central pivots n_p — and its deterministic boundary-row rule, at the operating point $\alpha = 0.7$ on 1 500 WISDM

windows (4 500 axis-series). The floating-point inverse is numerically *exact* there (MAE $< 10^{-6}$): the closed-form reconstruction carries no algorithmic error, so the entire GADF error reported elsewhere in this paper is 8-bit PNG quantisation (physical MAE 0.0158, Pearson 1.0000), not decoder failure. The boundary anchor never mis-identifies (0% failure across the 4 500 axis-series), because per-window min–max normalisation forces both amplitude extremes to be present, so the extreme-mean row is always a genuine boundary. We fix $\alpha = 0.7$ as the largest value below the branch-uniqueness threshold $1/\sqrt{2} \approx 0.7071$ (Section 4.4), which maximises the effective use of the 8-bit intensity range while keeping every pairwise angular difference inside the principal branch of arcsin.

Pivot-ensemble size. Averaging the reconstruction over the n_p most-central rows, rather than trusting a single pivot, is the second numeric choice. Table 11 sweeps n_p on the same PNG round trip at $\alpha = 0.7$: the pointwise error falls monotonically — more than halving from a single pivot (0.0353) to five (0.0158, the value used throughout the paper) and improving further to 0.0119 at $n_p = 9$ — at essentially constant cost (1.54–1.59 ms per axis-series), since each extra pivot is one more $O(N)$ pass over an already-loaded matrix. We fix $n_p = 5$ as a conservative default that captures most of the averaging gain; a larger ensemble would lower the reported GADF error further at negligible extra cost, but n_p is kept fixed so the headline numbers are not tuned on the evaluation windows.

Table 11: GADF pivot-ensemble sweep on the same 1 500 WISDM windows (4 500 axis-series) at $\alpha = 0.7$. n_p is the number of averaged central pivot rows; MAE is the 8-bit PNG round-trip error in physical units; time is per axis-series. The default used everywhere else in the paper is $n_p = 5$.

n_p	1	3	5	7	9
MAE (8-bit PNG)	0.0353	0.0206	**0.0158**	0.0134	0.0119
Time (ms/axis)	1.54	1.55	1.57	1.58	1.59

6.6. RP Decoder Ablation: Inverse Normalisation and Signed Orientation Selection

Table 12 separates correct RP round-trip recovery from the two algorithmic choices of the proposed decoder, on 100 stratified window-level fold-0 test windows. Decoding the stored image with the RP matrix's own range rather than the signal amplitude range (Step 1) is a *round-trip correctness precondition*: with the wrong range the reconstruction is broken rather than merely worse, so we report it only as a sanity-check row. The two genuine algorithmic choices are retaining the *full two-dimensional state geometry* instead of a scalar projection, and the *signed orientation selection* (Step 5) that fixes the residual reflection of that embedding. We also report EVR_2, the fraction of the multidimensional-scaling spectrum captured by the two leading positive dimensions — a direct diagnostic of whether the recovered distances still form a two-dimensional Euclidean geometry.

Table 12: RP decoder ablation on 100 stratified window-level fold-0 test windows; MAE (raw) and Pearson are averaged over the three axes, and EVR_2 is the fraction of the MDS spectrum in the two leading positive dimensions. All rows use the RP-range inverse normalisation except where noted.

Variant	MAE	Pearson	EVR_2
DIRECT-2D-S (proposed decoder)	**0.010**	**1.000**	0.947
with signal-range decode (uncorrected)	1.047	0.959	0.218
without signed orientation selection	2.033	0.794	0.947
Prior one-dimensional graph decoder (Jaccard)	2.177	0.918	0.730

The sanity-check row confirms the round-trip precondition: decoding with the signal amplitude range instead of the RP range raises the MAE from 0.010 to 1.047 and, more tellingly, collapses EVR_2 from 0.95 to 0.22, because the affine scale mismatch turns the recovered magnitudes into a matrix that is no longer two-dimensionally Euclidean, so multidimensional scaling can no longer place the states. The two algorithmic choices are each decisive in turn: removing the signed orientation selection leaves the reflection unresolved and raises the MAE to 2.033, and the prior one-dimensional graph decoder — which projected each state onto a single MDS axis and re-augmented the length with a per-window mean — reaches only MAE = 2.18 even with the corrected round-trip, confirming that recovering the full two-dimensional state geometry, not just a scalar projection, is what makes the signal recoverable.

6.7. MTF Decoder Ablation: Ordering and Metadata

The MTF decoder has two design choices we need to defend empirically: the bin-ordering algorithm (cosine-pivot heuristic vs Fiedler-vector spectral) and the bin count Q. We ablate both on the stratified window-level 3-fold protocol against the same six classifiers used in the rest of the paper.

Decoder ablation (at fixed $Q = 4$). We compare four decoder configurations on the same encoded MTF images (Table 13): the cosine-pivot ordering heuristic, Fiedler-vector ordering, Fiedler-vector ordering with stored bin-midpoint metadata, and Fiedler-vector ordering with both midpoint metadata and a sign-bit. The encoder is held constant. Bin midpoints (storing Q floats per axis, the median of each quantile bin) and a sign-bit (one bit per axis to disambiguate the mirror symmetry) are *not* part of the side-information contract used for the headline numbers; they characterize the achievable floor that any MTF decoder operating on top of the standard MTF image cannot beat.

Table 13: MTF decoder ablation at $Q = 4$, 100 stratified window-level fold-0 test windows ($T = 128$); pointwise MAE and Pearson averaged across the three accelerometer axes.

Decoder variant	MAE	Pearson
Cosine-pivot heuristic	6.493	+0.023
Fiedler ordering	5.060	−0.008
Fiedler + midpoint metadata	3.926	+0.013
Fiedler + midpoint + sign-bit	2.603	+0.465

The Fiedler ordering alone, with no extra metadata, drops the cosine-pivot pointwise MAE by 22% (6.49 → 5.06). Adding Q bin-median floats per axis (comparable in cost to the optional per-axis $(\min, \max)$ calibration used only for physical-scale recovery) drops it to 3.93; adding a single sign-bit per axis – which resolves the global reversal ambiguity of the inferred Fiedler ordering, though not the many-to-one quantisation and transition-profile ambiguities intrinsic to MTF – drops it further to 2.60. We use *Fiedler ordering without midpoint or sign-bit metadata* for every MTF number elsewhere in the paper because it keeps the side-information contract identical to the cosine-pivot heuristic, and characterizes what an MTF decoder can achieve without extra metadata beyond the standard optional per-axis physical-scale calibration.

Bin-count ablation across Q. The bin count Q is the parameter most often discussed in the MTF literature. We sweep $Q \in \{2, 4, 6, 8, 10, 16\}$ across the full stratified window-level 3-fold test set ($N = 8\,172$ pooled windows; approximately 2 724 per fold) and re-validate the six-classifier panel on each reconstruction, using the Fiedler decoder uniformly. Table 14 gives the result; the rows are sorted by Q to make the location of the minimum explicit.

Table 14: MTF bin-count ablation on the stratified window-level 3-fold protocol ($N = 8\,172$ pooled test windows). Pointwise MAE and Pearson are averaged across the three accelerometer axes; classifier accuracies are 3-fold means. The minimum MAE, maximum Pearson, and per-classifier optimum are highlighted in bold. The Fiedler-vector decoder is used uniformly; only Q varies.

Q	**MAE**	**Pearson**	KNN	SVM	RF	LSTM	GRU	CNN
2	4.11	+0.002	0.490	**0.705**	**0.762**	0.756	**0.771**	**0.774**
4	**3.95**	**+0.013**	**0.559**	0.697	0.757	**0.757**	0.765	0.773
6	4.01	+0.004	0.556	0.697	0.757	0.755	0.761	0.770
8	4.03	−0.001	0.539	0.695	0.753	0.757	0.764	0.765
10	4.07	+0.000	0.527	0.693	0.754	0.756	0.762	0.758
16	4.13	−0.014	0.520	0.697	0.755	0.757	0.762	0.753

Three observations matter:

1. **Pointwise fidelity is jointly maximised at $Q = 4$.** Both MAE and Pearson reach their optimum at $Q = 4$ (3.95, +0.013); every other choice of Q strictly degrades both metrics. The amplitude of the variation is modest (MAE moves in $[3.95, 4.13]$ across the swept range) because the Fiedler decoder is dominated by the intra-bin variance regardless of Q, but the location of the minimum is unambiguous.

2. **The amplitude-sensitive KNN classifier peaks at $Q = 4$.** KNN – the only classifier in our panel that measures local geometric similarity in the flattened sample space – improves by +6.9 percentage points going from $Q = 2$ to $Q = 4$ (0.490 → 0.559) and then monotonically degrades down to 0.520 at $Q = 16$. CNN, which extracts local temporal patterns through learned convolutional filters, is similar at $Q = 2$ and $Q = 4$ (0.774 vs 0.773) and decreases moderately for larger Q (to 0.753 at $Q = 16$). KNN, the classifier that most exploits fine-grained amplitude structure, is thus the one that clearly pays for under- or over-quantisation.

3. **Coarse-pattern classifiers are nearly insensitive to Q.** SVM, Random Forest, LSTM and GRU vary by at most 0.012 accuracy across the entire $Q \in \{2, \ldots, 16\}$ sweep, with their (statistically negligible) optimum located at $Q = 2$. These four classifiers absorb most of the discriminative signal through bin-transition counts rather than through quantised amplitude shape, which is why they tolerate the coarsest quantisation. Their indifference to Q also explains why MTF as a representation is robust to the choice of Q: in the regime where most of the activity signal lives in the transition structure (high-ceiling neural classifiers), MTF's pre-quantisation step is not the bottleneck.

Taken together, the two ablations defend the $Q = 4$ default empirically: it is the unique value that minimises pointwise MAE, maximises pointwise Pearson, and maximises the amplitude-sensitive KNN accuracy (CNN being essentially flat, within 0.001 of its $Q = 2$ value), while costing at most 0.012 accuracy on the four classifiers that are essentially flat across Q.

7. Discussion

The four encodings have now been measured under the same WISDM protocol – $N = 128$ windows, full stratified window-level 3-fold evaluation ($N = 8\,172$ pooled test windows; $\sim$2 724 per fold), six classifiers fit *once per fold* on the originals and evaluated on every reconstruction – so the comparison no longer mixes one fully evaluated method against partially evaluated ones and is no longer restricted to a single split. We organise the discussion around four lenses (visual reconstruction, pointwise error, polynomial-trend similarity, and downstream classification) and ask, in turn, where the four encodings sit on each lens and whether the rankings agree. Table 15 states the role we assign to each operator after this round of evaluation; the compact classifier summary that underpins it, side by side for the window-level and subject-out protocols, is Table 16, the single most useful read of the comparison.

Table 15: Interpretation of each encoding after the unified $N = 128$ evaluation.

Method	**Strength**	**Main Limitation**	**Empirical role**
RP	Recurrence geometry and temporal morphology	Needs the RP-range inverse normalisation and resolution of the $O(2)$ MDS ambiguity via the state overlap and signed convention	Near-original on every lens: near-lossless pointwise (MAE ≈ 0.015) and matches original-signal classification
GASF	Pointwise reconstruction through diagonal information (closed form)	Branch ambiguity outside $[0, \pi/2]$; min–max needed only for physical units	Near-original on every lens; tied with GADF as the two near-original decoders
GADF	Directional angular differences with $O(N^2)$ boundary-anchored angular inverse ($O(N)$ reconstruction after $O(N^2)$ row statistics)	Branch, sign convention, and angular anchor; min–max only for physical units	Near-original decoder, statistically tied with GASF in classification and marginally tighter on pointwise error; closed-form once the anchor row is fixed
MTF	Coarse transition dynamics, partly robust to quantisation; spectral Fiedler-vector ordering halves the cosine-pivot pointwise MAE	Intra-bin magnitude loss and global mirror ambiguity (a single sign bit would resolve it but is not stored here)	The only lossy decoder: last on every lens, significantly outranked by RP, GASF and GADF on all six classifiers; a compact classifier-oriented representation rather than a faithful inverse (Section 7.6)

Table 16: Stratified window-level vs grouped subject-out accuracies (mean across 3 folds) — the compact summary that the rest of this section unpacks. The grouped subject-out protocol holds out 16–18 subjects per fold with zero overlap, so it is the stricter subject-generalisation test. The near-original tier — RP, GASF and GADF — is preserved under both sampling schemes: all three match the original-signal accuracy on every classifier. MTF is the single decoder that degrades, and it is significantly outranked by RP, GASF and GADF on all six classifiers under both protocols. Absolute accuracies are lower under grouped subject-out validation because the model has not seen any window from the test subjects.

	Orig.		**GASF**		**GADF**		**RP**		**MTF**	
Classifier	Window	Subject-out	Window	Subject-out	Window	Subject-out	Window	Subject-out	Window	Subject-out
GRU	0.923	0.811	0.923	0.811	0.923	0.811	0.923	0.811	0.765	0.702
KNN	0.642	0.504	0.642	0.504	0.643	0.505	0.642	0.504	0.559	0.465
CNN	0.907	0.796	0.907	0.796	0.907	0.796	0.901	0.794	0.773	0.686
Random Forest	0.891	0.786	0.891	0.787	0.892	0.787	0.892	0.786	0.757	0.689
Support Vector Machine	0.801	0.738	0.801	0.738	0.801	0.738	0.802	0.739	0.697	0.665
LSTM	0.869	0.744	0.868	0.744	0.869	0.744	0.869	0.744	0.757	0.672

Complete per-class precision, recall, and F1-score results are provided in Appendix B.

7.1. Visual Reconstruction Trade-offs

Figures 15–18 show the four reconstructions overlaid on the original triaxial window. GASF (Fig. 16) is visually indistinguishable from the original on all three axes, including high-frequency wiggles, confirming that the diagonal half-angle rule recovers the normalised shape and that the optional physical-scale calibration restores the native amplitude. GADF (Fig. 17) is likewise visually indistinguishable from the original, including the high-frequency wiggles; once the branch, anchor and sign convention are fixed, the closed-form inverse leaves only 8-bit PNG quantisation

error, matching the near-original pointwise numbers of Section 7.3. RP (Fig. 15) recovers the full morphology of the window: with the corrected inverse normalisation the deterministic classical-MDS embedding and the signed-RP orientation rule leave only residual quantisation-level error, so the reconstruction overlays the original much as the Gramian fields do, matching the near-original pointwise numbers of Section 7.3. MTF (Fig. 18) tracks coarse structure but reduces the signal to a small number of plateaus, the unavoidable consequence of representing every sample by a $Q = 4$ state label.

7.2. Null-Corrected RP/GADF Regional Overlap

The visual proximity between RP and GADF is easy to overstate from a single example, so we add a diagnostic test that deliberately separates spatial region alignment from marginal colour statistics. On the full set of non-overlapping smartwatch-accelerometer WISDM windows (window length $N = 128$), we encode the same signal as signed RP and GADF, align the matrices on the common 127×127 support, and quantise each image channel into B visual intensity ranges. For every matched pair $(\mathrm{RP}_i, \mathrm{GADF}_i)$, we compare the quantised regions against a conservative null pair $(\mathrm{RP}_i, \mathrm{GADF}_j)$, where $j \neq i$ is drawn from the same activity class. The primary statistic is a weighted regional Jaccard score, $\sum_b |R_b \cap G_b| / \sum_b |R_b \cup G_b|$, averaged across the grayscale view and the three RGB/axis channels. We also compute a distribution-only overlap that ignores pixel position and a boundary-F1 score between quantised region edges.

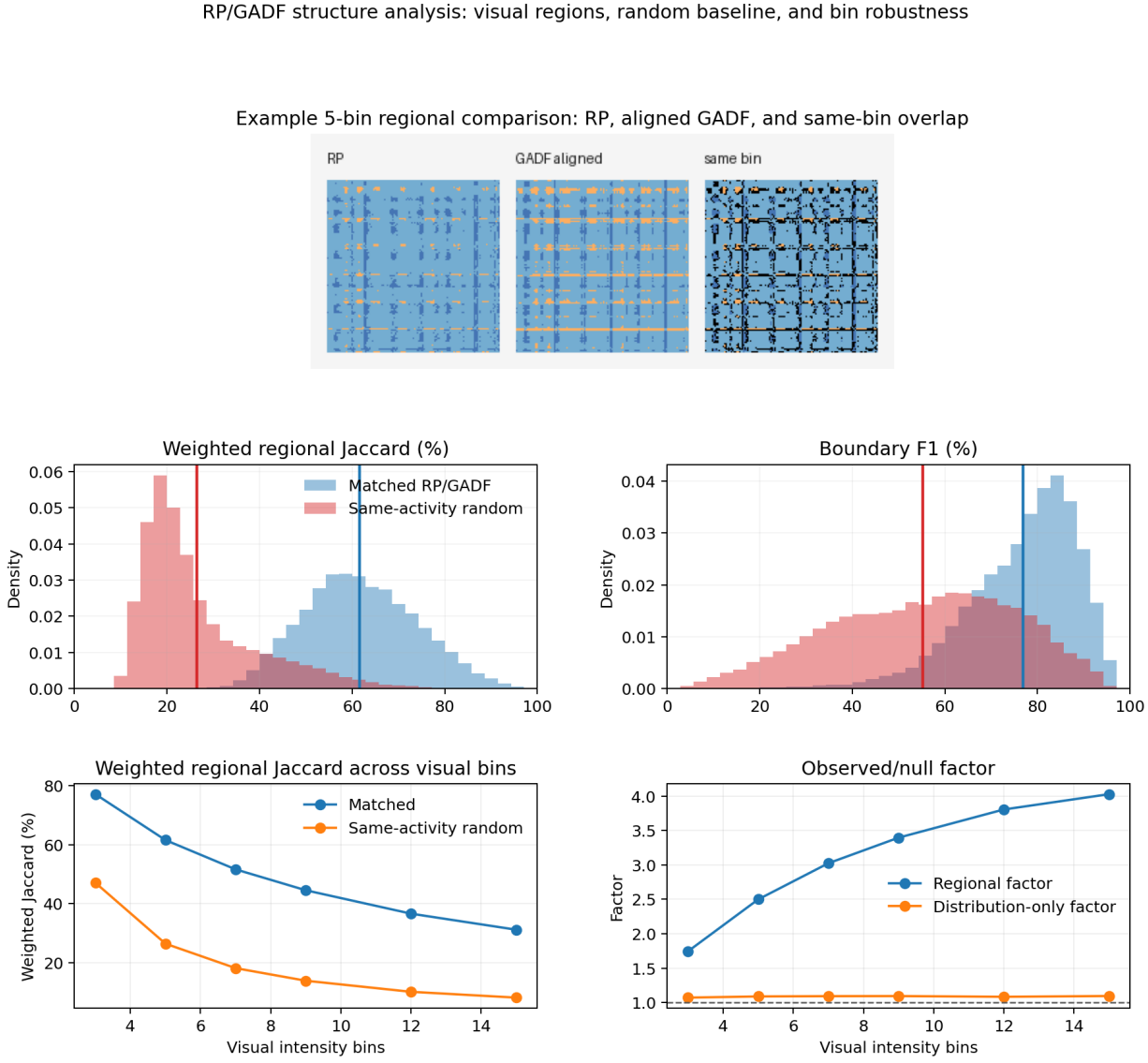


Figure 22: Null-corrected RP/GADF structural-overlap diagnostic. The top panel shows one 5-bin visual comparison: quantised RP, aligned quantised GADF, and pixels landing in the same bin. The middle histograms compare matched RP/GADF pairs against same-activity random pairings. The bottom row shows that matched regional overlap remains above the null baseline across independent visual quantisation levels $B \in \{3, 5, 7, 9, 12, 15\}$, while the distribution-only factor stays close to one.

Table 17: Null-corrected RP/GADF weighted regional overlap across visual intensity bins. Values are averaged over grayscale and the three axis channels. The null baseline pairs each RP image with a GADF image from a different window of the same activity class. Jaccard values are percentages; the Matched/null factor is the mean over the four channels of the per-channel matched/null ratio, so it is not the ratio of the two displayed column means.

Visual bins B	Matched Jaccard	Same-activity null	Matched–null gain	Matched/null factor
3	77.16	47.11	+30.05	1.75×
5	61.57	26.43	+35.14	2.50×
7	51.64	18.14	+33.50	3.02×
9	44.55	13.87	+30.68	3.40×
12	36.62	10.12	+26.50	3.81×
15	31.17	8.14	+23.03	4.03×

Table 17 and Figure 22 support a narrow but useful claim: RP and GADF are not equivalent encodings, yet their coarse regions are substantially more aligned when they come from the same signal window than when activity label alone is controlled. As B increases, the absolute Jaccard necessarily decreases because the region match becomes stricter, but the matched/null factor rises from 1.75× to 4.03×. This is exactly the pattern expected if the shared structure is spatial rather than a by-product of broad intensity histograms. The distribution-only factor stays near 1.07×–1.09× across the same sweep, so marginal colour occupancy explains only a small part of the effect. Boundary F1 gives the same sign of evidence at coarse quantisation, but becomes less diagnostic at high B because every image develops many region edges; for that reason we treat weighted regional Jaccard as the primary overlap statistic.

7.3. Reconstruction Error Analysis

The pointwise table (Table 7, over all 8172 windows; per-axis breakdown in Appendix A) confirms the visual ordering. GASF attains a global MAE of 0.0205 ± 0.0199 and a Pearson correlation of 0.99988 ± 0.00014; the residual error is at the level of 8-bit image quantisation (a 1/255 step in the normalised stored intensity, before denormalisation), which rules out a structural failure of the diagonal inverse. GADF reaches MAE $= 0.0157 \pm 0.0167$ and Pearson 0.99999 ± 0.00001 – effectively tied with GASF and roughly two orders of magnitude below MTF: once the arcsin branch, the boundary anchor and the sign convention are fixed, the closed-form inverse is exact up to the 8-bit PNG quantisation of the stored image, so its residual error is likewise at the 8-bit quantisation level, on par with GASF's. RP yields MAE $= 0.0148 \pm 0.0170$ and Pearson 0.99998 ± 0.00005; once the inverse normalisation is corrected its error sits at the same 8-bit-quantisation level as the two Gramian decoders — in fact marginally the lowest global MAE of the three — so RP joins GASF and GADF as a near-original pointwise decoder. All three error distributions are right-skewed, with medians well below their means (RP 0.0091, GADF 0.0112, GASF 0.0158; Table 7), so the typical window is reconstructed even more accurately than the mean suggests. MTF under the Fiedler-vector decoder of Section 6.7 reaches MAE $= 3.94 \pm 4.25$, MSE $= 55.43 \pm 113.46$ and a Pearson correlation that averages to only 0.014 with a wide per-window spread (global std 0.489, rising to ± 0.85 per axis; Appendix A). That spread is the diagnostic of the residual mirror ambiguity: per window, MTF correlates either strongly positively or strongly negatively with the original because the recovered amplitude order can land on the wrong side of the global flip, and the two outcomes cancel in the mean.

The pointwise orderings are statistically significant under paired, distribution-free testing. Wilcoxon signed-rank on the per-window MAE pooled across the 3 stratified window-level folds ($N = 8\,172$ paired windows) rejects equality between every pair of encoders ($p_{\text{Holm}} \leq 1.2 \times 10^{-90}$, and $< 10^{-300}$ for five of the six pairs). Four of the six effect sizes $|r| = Z/\sqrt{N}$ are in the large-effect regime under the Rosenthal convention: the three comparisons that pit a near-lossless decoder against MTF (RP-, GASF- and GADF-vs-MTF, $|r| = 0.866$ throughout, reflecting the two-orders-of-magnitude gap to the quantised Markov decoder) plus RP-vs-GASF ($r = -0.615$). The remaining two are smaller: GASF-vs-GADF is a medium effect ($r = +0.457$) and RP-vs-GADF a small effect ($r = -0.223$), both reflecting sub-quantisation differences within the near-lossless trio. The sign of r recovers a single consistent per-window ranking, RP $\succ$ GADF $\succ$ GASF $\succ$ MTF, with RP attaining the lowest per-window MAE by a small margin and all three reconstruction decoders far above MTF.

This window-level ordering is not an artefact of the large window count. Repeating the paired Wilcoxon at the subject level — collapsing each subject to a single mean-MAE value so that the $N = 51$ subjects, rather than the 8 172 windows, are the independent units — preserves the two features that matter: MTF stays two orders of magnitude above the three near-lossless decoders (per-subject mean MAE 3.71 for MTF against ≈ 0.02 or below for RP, GADF and GASF alike), and the GADF-vs-GASF gap again favours GADF (per-subject mean MAE 0.0148 vs 0.0199; the subject-clustered Wilcoxon rejects equality in GADF's favour). The subject-clustered test therefore confirms that MTF's amplitude loss and the marginal GADF-over-GASF advantage are per-subject effects, not pseudoreplication artefacts of the 8 172 windows, while RP, GADF and GASF remain a tightly clustered near-lossless group at the 8-bit-quantisation floor.

7.4. Polynomial Trend Comparison

Compressing each min–max-normalised window into a global cubic-polynomial coefficient vector (a_3, a_2, a_1, a_0) removes most sample-level noise and asks only whether the broad shape is right. The result (Table 8, computed with the standardized-coefficient distance metrics defined in Section 5.3.3) places the three near-lossless decoders together at the top: GADF, GASF and RP are effectively tied (cosine similarity 1.00000, 0.99975 and 0.99991; standardized Euclidean coefficient distance 0.00070, 0.00081 and 0.00081), while MTF trails well behind (cosine 0.90610). Two observations matter: (i) RP's polynomial-trend similarity is now on par with the Gramian fields, consistent with the corrected decoder recovering both the shape and the amplitude of the window, and (ii) MTF still retains a usable rough trend (cosine 0.90610) even though its pointwise correlation collapses to zero, a property that the Fiedler-vector spectral decoder of Section 6.7 leverages into a compact classifier-oriented representation (Section 7.5).

7.5. Classification Performance

The six per-classifier tables show that RP, GASF and GADF all reproduce the original-signal accuracy on every classifier, while MTF is the only reconstruction whose accuracy drops, and they identify where the reconstructions are good enough to be downstream-equivalent to the raw signal.

GASF reconstructions show no statistically significant degradation relative to the originals across all six classifiers ($|\Delta\text{acc}| \leq 0.001$ on the 3-fold mean for KNN, SVM, LSTM, Random Forest, CNN, and GRU). The paired McNemar test [20] confirms this on the pooled $N = 8\,172$ windows: in every one of the six Original-vs-GASF comparisons the raw p-value lies between 0.45 and 1.00 and the Holm-adjusted value is 1.00, so we do not detect a paired classification difference under this protocol (24-test family, $\alpha = 0.05$). This is not a formal equivalence test: failing to reject a difference is not, by itself, proof of equivalence or information preservation. Combined with the near-zero pointwise error, it supports the narrower claim that GASF is empirically near-equivalent for the evaluated WISDM task under controlled encoding settings and, for the reported physical-unit comparison, per-window scale calibration. The pre-specified-margin equivalence test this calls for — a paired TOST on Δaccuracy — is reported below at a ± 0.02 margin and confirms the equivalence at the subject level.

GADF shows no detected degradation relative to the originals on any classifier ($|\Delta\text{acc}| \leq 0.001$ on the 3-fold mean; on Random Forest GADF is even 0.001 above the original). The paired McNemar test on the pooled $N = 8\,172$ windows fails to reject Original-vs-GADF in all six classifiers ($p_{\text{Holm}} = 1.0$ throughout), and equally fails to separate GADF from GASF ($p_{\text{Holm}} = 1.0$ in all six GASF-vs-GADF pairs): no statistically detectable classification difference is observed between the two Gramian decoders or from the raw signal. The same McNemar family confirms that GADF significantly beats MTF on every classifier ($p_{\text{Holm}} < 10^{-20}$), while it shows no detectable difference from RP on any classifier ($p_{\text{Holm}} = 1.0$ in all six GADF-vs-RP pairs): GADF, GASF and RP form a single near-original tier, jointly separated from MTF. As with GASF, this is a difference test, not a formal equivalence test: absence of significance is reported as no detected degradation, not proof of equivalence.

RP matches the original-signal accuracy on every classifier (3-fold window-level means within 0.006 of the original — KNN 0.642, SVM 0.802, Random Forest 0.892, LSTM 0.869, GRU 0.923, CNN 0.901, the largest gap being CNN's 0.901 vs 0.907). The paired McNemar test fails to reject Original-vs-RP in all six classifiers ($p_{\text{Holm}} = 1.0$ throughout). It is, however, decisively separated from MTF: RP significantly outranks MTF on every classifier (p_{Holm} from 1.8×10^{-68} on KNN to 4.5×10^{-226} on GRU, all in RP's favour), so the corrected RP reconstruction is a near-original decoder on the classifier lens rather than a lossy one. As with the Gramian decoders, this window-level McNemar is a difference test; the subject-clustered equivalence analysis below supplies the corresponding equivalence test.

Subject-clustered analysis and a formal equivalence test. The McNemar tests above pool windows and treat them as independent, which the intra-subject correlation of activity windows makes mildly anti-conservative (Section 7.9). We therefore repeat the classification comparison with the $N = 51$ subjects — not the 8 172 windows — as the independent units, and add the formal equivalence test that a difference test cannot provide. For every classifier we compute the per-subject accuracy of each source and run (i) a subject-clustered Wilcoxon signed-rank test between sources and (ii) a paired two-one-sided-tests (TOST) equivalence test on the per-subject Δaccuracy at a pre-specified margin of ± 0.02 (two percentage points). The subject-clustered Wilcoxon detects no accuracy difference between the original and any of RP, GASF or GADF, nor among those three, on any classifier ($p_{\text{Holm}} = 1.0$ throughout), while MTF is separated from all of them ($p_{\text{Holm}} < 10^{-5}$). The TOST is positive exactly where the difference test is null: RP, GASF and GADF are each statistically *equivalent* to the original — and to one another — within the two-point margin on all six classifiers (Holm-adjusted TOST $p < 10^{-21}$; per-subject mean $|\Delta\text{acc}| \leq 0.0011$), whereas MTF is equivalent to none (per-subject Δacc between 0.08 and 0.16). Treating subjects as the unit of analysis therefore upgrades the earlier "no detected difference" to positive equivalence within a two-point margin for the three near-lossless decoders, and confirms the MTF separation, without relying on the window-independence assumption; the per-(classifier, source) tables are in the accompanying statistical report.

MTF, under the Fiedler-vector decoder of Section 6.7, is the only reconstruction that degrades classification: it loses 0.11–0.16 accuracy on the high-ceiling classifiers, 0.10 on SVM, and 0.08 on KNN (3-fold means). Two findings about MTF deserve to be highlighted because they make quantitative and statistically signed the standard intuition that MTF is "good for classification, bad for reconstruction".

First, MTF is significantly outranked by all three near-lossless decoders on every classifier. The paired McNemar test on the pooled $N = 8\,172$ window-level sample rejects RP-vs-MTF equality on all six classifiers, every one in RP's favour (p_{Holm} from 1.8×10^{-68} on KNN to 4.5×10^{-226} on GRU), and the GASF-vs-MTF and GADF-vs-MTF families reject at the same magnitudes. MTF is therefore below RP, GASF and GADF on the classifier lens without exception, mirroring its position on the pointwise and polynomial-trend lenses. We read this as a direct consequence of the $Q = 4$ quantisation: MTF discards within-bin amplitude and keeps only quantile-ordered transition statistics, which suffices for a coarse activity decision but costs accuracy relative to the three near-lossless reconstructions.

Second, the four lenses now agree. On the pointwise, polynomial-trend and classifier lenses alike the ordering is $\{\text{RP}, \text{GASF}, \text{GADF}\} \succ \text{MTF}$: the three near-lossless decoders are jointly indistinguishable from the original and from each other, and MTF is the single lossy outlier on every lens. There is no rank inversion between reconstruction fidelity and downstream utility — the earlier apparent trade-off was an artefact of the uncorrected RP inverse. Per-class behaviour is consistent with MTF's quantisation: classes for which the source signal is amplitude-monotonic (sitting, standing) survive the $Q = 4$ binning reasonably well, and classes that depend on amplitude shape (jogging, stairs) lose precision but retain recall through the preserved transition order.

7.6. Four-lens Convergence

Aggregating across the four lenses we observe a clean convergence. RP, GASF and GADF sit together as a near-original tier — indistinguishable from one another and from the raw signal in classification (McNemar $p_{\text{Holm}} = 1.0$ in every Original-vs-RP, Original-vs-GASF, Original-vs-GADF and mutual encoder pair) and clustered at the 8-bit-quantisation floor on pointwise error — while MTF sits below them on every lens. The three near-lossless decoders are separated only by sub-quantisation differences on per-window MAE (a small RP-vs-GADF effect, $r = -0.223$, and a medium GASF-vs-GADF effect, $r = +0.457$, both far below the maximal $|r| = 0.866$ that separates each of them from MTF). The proximity between RP and GADF on directional pairwise structure – developed in Section 7.7 – is visible in every lens: GADF is strictly antisymmetric, while signed RP is a zero-diagonal directional distance matrix whose external sign rule approximates a related pairwise-difference structure without guaranteeing exact antisymmetry.

The one clear separation is between the near-lossless trio and MTF, and it is consistent across every lens. On the pointwise lens each of RP, GASF and GADF is orders of magnitude closer to the original than MTF (MAE = 0.0148, 0.0205 and 0.0157 vs 3.94; Wilcoxon $|r| = 0.866$ for all three against MTF); on the polynomial-trend lens the same three cluster near cosine similarity 1.0 while MTF trails at 0.90610; and on the classifier lens all three match the original on every classifier while MTF is significantly outranked throughout. The lenses are therefore neither redundant nor contradictory here: they agree that RP, GASF and GADF are reversible substrates and that MTF is a lossy, classifier-oriented representation. For a reconstruction-faithful augmentation pipeline the three near-lossless decoders are interchangeable substrates, $\{\text{RP}, \text{GASF}, \text{GADF}\} \succ \text{MTF}$; MTF remains useful only where a compact quantile-transition summary, rather than a faithful inverse, is what the downstream consumer needs.

7.7. Structural relationship between signed RP and GADF

Operator-level comparison of pairwise structure. Before discussing the first-order amplitude expansion that the rest of this section relies on, it is useful to put signed RP and GADF in a common notation without claiming equality of their output spaces. Both can be written as pairwise functions of lifted scalar samples,

$$M_{ij} \;=\; \omega\big(\xi(x_i),\, \xi(x_j)\big),$$

where $\xi : \mathbb{R} \to \mathbb{R}^2$ is a two-dimensional lift of the scalar sample and $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is the pairwise operator associated with the encoding. The two methods differ in both the lift and the operator:

- RP under the Lu–Tong convention uses the adjacent-state embedding $\xi_{\text{RP}}(x_k) = (x_k,\, x_{k+1})$ and a signed Euclidean distance, $\omega_{\text{RP}}(u, v) = \sigma(u, v)\|u - v\|_2$, where the sign is assigned *externally* by the projection onto the reference direction $v_\star = (1, 1)^\top/\sqrt{2}$. With the thresholded sign rule used here, $\sigma(u, v)$ is not guaranteed to satisfy $\sigma(u, v) = -\sigma(v, u)$ for every pair, so the resulting matrix is directional but not necessarily antisymmetric.
- GADF uses the polar lift $\xi_{\text{GADF}}(\tilde{x}_k) = (\tilde{x}_k,\, \sqrt{1 - \tilde{x}_k^2}) = (\cos\phi_k, \sin\phi_k)$ and the operator $\omega_{\text{GADF}}(u, v) = \det[v \mid u] = u_2v_1 - u_1v_2 = \sin(\phi_i - \phi_j)$, the signed area of the parallelogram spanned by the two lifted points, whose sign is *intrinsic* to the operator and matches G^{D}_{ij} in Eq. (8).

The important consequence is therefore an analogy, not an identity. GADF lies *within* the $\binom{N}{2}$-dimensional space of antisymmetric zero-diagonal matrices by construction, because the determinant changes sign when its arguments are swapped; more specifically, its factorisation $G^{\mathrm{D}} = \mathbf{s}\,\mathbf{c}^{\top} - \mathbf{c}\,\mathbf{s}^{\top}$ (with $s_i = \sin\phi_i$, $c_i = \cos\phi_i$) shows it occupies only a rank-≤ 2 subvariety of that ambient space, not the whole of it. Signed RP has a zero diagonal and carries directional pairwise-distance information, but the Lu–Tong sign rule used in this paper does not force exact antisymmetry for nearly orthogonal displacements. The first-order expansion below quantifies why the two images can nevertheless look and behave similarly, while the structural differences between the two operators — GADF's intrinsic sign and its closed-form one-pass inverse, versus signed RP's externally assigned sign and its MDS-based reconstruction — follow from the determinant-based operator rather than from an exact shared matrix space.

A structural reading of why RP and GADF look (and behave) so similarly. Beyond the empirical ranking, the proximity between RP and GADF deserves a structural reading: their visual resemblance and their adjacent positions in the four-lens ranking are not coincidental. GADF produces an antisymmetric zero-diagonal matrix, while signed RP produces a zero-diagonal directional distance matrix with related pairwise-difference content. For smooth signals whose local increments vary slowly (so that $x_{i+1} - x_i \approx x_{j+1} - x_j$ over the compared indices), both reduce, in first order, to functions of the pairwise-difference matrix $\{x_i - x_j\}_{i,j}$. The signed RP under the Lu–Tong embedding satisfies

$$R_{ij} \;\approx\; \sigma(i,j)\,\sqrt{2}\,|x_i - x_j|,$$

where $\sigma(i,j)$ is the external orientation flag computed from the projection of $s_i - s_j$ onto $v = (1,1)^{\top}/\sqrt{2}$. The GADF entry, expanded around a common normalized amplitude $\bar{x}$, satisfies

$$G^{\mathrm{D}}_{ij} \;=\; \sin(\phi_i - \phi_j) \;\approx\; \frac{\bar{x}_j - \bar{x}_i}{\sqrt{1-\bar{x}^2}},$$

because $d(\arccos x)/dx = -1/\sqrt{1-x^2}$, with both the samples $\bar{x}_i, \bar{x}_j$ and the expansion point $\bar{x}$ taken in the restricted $[-\alpha, \alpha]$ normalization of the GADF decoder. The sign of the GADF term is intrinsic to the trigonometric form rather than imposed externally. Thus, both expressions are linear in pairwise amplitude differences at first order and both are zero on the diagonal, but only GADF is strictly antisymmetric under the stated construction. What makes the restricted-branch GADF decoder closed-form is that it (i) makes the sign part of the encoding rather than a post-hoc convention, and (ii) restricts the angular range to $[\arccos(\alpha), \arccos(-\alpha)]$ with $\alpha = 0.7$, a near-maximal symmetric choice below $1/\sqrt{2}$ that keeps every $\sin(\phi_i - \phi_j)$ inside the principal branch of arcsin. In other words, GADF and signed RP are structurally adjacent parametrizations of directional pairwise differences: GADF makes the inverse map explicit in one closed-form pass under the branch restriction, while signed RP recovers the same directional structure through classical MDS with a signed-orientation step. This reading clarifies why GADF is the natural Gramian counterpart to signed RP — the two are closer to each other than to either GASF or MTF, and both resolve the orientation ambiguity of the pairwise-difference representation, GADF intrinsically through sin and signed RP through its external sign convention.

Practical implication: RP, GASF and GADF are three near-lossless inverses of a shared directional-difference structure. The structural analogy above has a direct consequence for interpretation. The visual resemblance between the RP and GADF images is not a coincidence of pixel statistics — it follows from the first-order relations $R_{ij} \propto \sigma(i,j)|x_i - x_j|$ and $G^{\mathrm{D}}_{ij} \propto (\bar{x}_j - \bar{x}_i)/\sqrt{1-\bar{x}^2}$ derived above, and the null-corrected regional-overlap diagnostic of Section 7.2 confirms that matched RP/GADF windows share more spatial region structure than same-activity random pairings. GASF encodes the symmetric, amplitude-bearing part of the Gramian (diagonal $\cos(2\phi_i)$); GADF encodes the antisymmetric, difference-bearing part ($\sin(\phi_i - \phi_j)$); signed RP encodes a directional distance structure with additional sensitivity to the local derivative through the (x_i, x_{i+1}) pair embedding. Our empirical numbers confirm this analytical picture: RP, GASF and GADF all invert near-losslessly and are statistically indistinguishable from the original on every classifier, so all three serve as reversible substrates for the controlled reconstruction problem studied here. GADF has the practical advantage of a closed-form, one-pass inverse through the determinant operator, whereas RP additionally exposes recurrence geometry and therefore remains the natural choice when recurrence quantification, dynamical invariants or line-structure statistics are themselves the object of study.

7.8. Scope of Claims

The empirical scope of the conclusions is the full stratified window-level 3-fold protocol and the grouped subject-out 3-fold protocol on the smartwatch accelerometer stream of WISDM, with windows of $N = 128$ at 20 Hz and five activities ($N = 8\,172$ pooled paired test windows in each protocol). Within this scope, RP, GASF and GADF are all near-original decoders: they show no statistically detectable classification difference from the raw signal or from each other (paired McNemar $p_{\mathrm{Holm}} = 1.0$ throughout), and their pointwise errors sit together at the 8-bit-quantisation floor (global MAE 0.0148, 0.0205 and 0.0157 respectively, over all 8172 windows), with GADF marginally the tighter

of the two Gramian fields and RP the lowest of the three by a small per-window margin. Per-fold accuracy standard deviations are bounded by 0.03 under the window-level split and by 0.05 under grouped subject-out validation. The ranking is consistent across all four lenses: $\{\mathrm{RP}, \mathrm{GASF}, \mathrm{GADF}\} \succ \mathrm{MTF}$ on visual reconstruction, pointwise error, polynomial trend and downstream classification alike, with MTF the single lossy decoder — significantly outranked by each of the three near-original decoders on every classifier under both protocols (Section 7.6). No reconstruction improves on the raw signal: the three near-original decoders match it and MTF falls below it; we report empirical near-equivalence for this task rather than formal information preservation. MTF, under the spectral decoder of Section 6.7, retains value as a compact, classifier-oriented representation rather than a faithful inverse, so a practical recommendation should still name the downstream consumer wherever MTF is a candidate. The broader generative-augmentation claim is not addressed by these numbers: every decoder in this paper assumes that the image is the deterministic output of a known encoder with known encoder settings (the quantisation interval and sign convention); RP additionally requires the stored per-axis RP-matrix range even for its normalised reconstruction, and per-axis amplitude metadata are required only for physical-scale recovery. A diffusion or GAN that synthesises an image without storing those values would have to either learn them implicitly or be paired with a calibration step before any of these decoders can produce a physically plausible signal.

7.9. Threats to Validity

Several limitations bound the interpretation of these results and should be read alongside the rankings above.

- **The window-level protocol is a within-distribution split.** Its grouping key is effectively unique per window, so it measures robustness to reconstruction-induced distribution shift rather than generalisation to unseen subjects; the grouped subject-out (subject-disjoint, not strict leave-one-subject-out) numbers are the conservative estimate and the ones relevant to deployment.
- **Metadata dependence is conditional on the encoding and the output domain.** The Gramian and MTF decoders recover their *normalised* reconstruction from the image plus known encoder settings (the GADF sign convention and α, the MTF quantile count); RP additionally requires the stored per-window RP-matrix range $[R_{\min}, R_{\max}]$ for its normalised reconstruction. The stored per-axis amplitude min–max is needed *only* to map any normalised series back to physical accelerometer units (Section 4.1). The encoder settings, and physical-scale metadata when physical units are wanted, are unavailable for arbitrary generated images, which is why the work is framed as controlled reconstruction rather than generative augmentation.
- **GADF anchor and quantisation.** The angular boundary row is identified by a deterministic extreme-mean rule that is exact in the noise-free restricted regime; the ablation of Section 6.5 finds a 0% anchor-misidentification rate on WISDM, but this is dataset-dependent (near-constant windows have no reliable boundary), its robustness under perturbed or generated images remains to be evaluated, and 8-bit PNG storage remains the dominant error source.
- **Approximate McNemar homogeneity.** The pooled 2×2 table aggregates discordant pairs scored by three separately-trained per-fold models, so its p-values are mildly anti-conservative; a per-fold or Cochran–Mantel–Haenszel test would be stricter. This does not affect the substantive conclusions: the subject-clustered classification analysis of Section 7.5, which treats the 51 subjects rather than the windows as the units and so avoids the pooling assumption entirely, reaches the same conclusions (no detected difference among the three near-lossless decoders and their separation from MTF).
- **Intra-subject dependence.** Test windows from the same subject are not independent, so the window-level $p < 10^{-200}$ values overstate the effective sample size. The subject-clustered Wilcoxon ($N = 51$ subjects) in Section 7.3 confirms the separation of the three near-lossless decoders from MTF and the marginal GADF-over-GASF ordering at $p \leq 7.4 \times 10^{-10}$; the analogous subject-clustered analysis of the classifier comparisons (Section 7.5) likewise holds at the subject level — no detected accuracy difference among the three near-lossless decoders or from the original, and separation from MTF at $p_{\mathrm{Holm}} < 10^{-5}$.
- **Equivalence is margin-relative.** The classification equivalence of RP, GASF and GADF — to the original and to one another — is established by a paired TOST on per-subject Δaccuracy at a pre-specified ± 0.02 margin (Section 7.5), not merely by a failure to reject a difference. That margin is a modelling choice; it is far wider than the observed per-subject $|\Delta\mathrm{acc}| \leq 0.001$, so tighter margins would still pass, but the statement is inherently margin-relative and is an equivalence of downstream classification accuracy, not a proof of information-theoretic preservation.
- **Single dataset and protocol.** All numbers come from the WISDM smartwatch accelerometer stream, five activities, $N = 128$ windows at 20 Hz; the rankings need not transfer to other sensors, sampling rates, activity sets, or window lengths.

7.10. Reproducibility

All experiments use the WISDM smartwatch accelerometer stream [14] segmented into non-overlapping $N = 128$ windows at 20 Hz (8 172 triaxial windows). The pipeline runs on Python $\geq 3.10, < 3.12$ with TensorFlow 2.11–2.15 for the neural classifiers (CNN, GRU, LSTM) and scikit-learn for KNN, SVM and Random Forest. Two design choices keep the reconstruction reproducible: the RP decoder uses classical (Torgerson) MDS, which is a deterministic eigendecomposition rather than a randomly-initialised iterative solver, and the parallel reconstruction carries an explicit sample index so decoded windows stay aligned with their labels regardless of completion order. Both the stratified window-level and grouped subject-out 3-fold partitions are produced by a fixed `StratifiedGroupKFold` split. The encoders, the four decoders, the six-classifier panel, the statistical-test script, and the table-emission and verification scripts that regenerate every table in this paper from the stored results are provided in the accompanying code repository.

8. Conclusion

We have presented a controlled, symmetric evaluation of four image-based encodings – RP, GASF, GADF, MTF – as inverse mappings from image to triaxial time series. Three of the four reconstruction pipelines used in this evaluation are themselves new operational contributions of this work and, to the best of our knowledge, have no prior fully specified counterpart in the literature reviewed here: the signed-RP-aware decoder of Section 4.2, which reconstructs the scalar sequence from the *continuous signed* Lu–Tong RP of overlapping states — a case not addressed by Hirata, Horai, and Aihara [7], whose graph-geodesic reconstruction of the original series from unsigned recurrence plots leaves a multidimensional-scaling reflection ambiguity — by using the signed Lu–Tong definition as an in-loop constraint that resolves that ambiguity; the restricted-branch, multi-pivot, boundary-anchored GADF decoder of Section 4.4, which turns the previously stated but under-specified invertibility of GADF on restricted angular ranges into an explicit operational decoder with branch control, angular anchoring, fixed sign convention, optional min–max calibration for physical units, and empirical evaluation; and the Fiedler-vector spectral MTF decoder of Sections 4.5 and 6.7, which reconstructs a representative compatible quantile trajectory rather than a bijective inverse of MTF. The fourth pipeline, the GASF decoder, is the only one that has direct precedent in the literature [11] and is kept in the comparison as a tightly controlled prior-art baseline. The four encoders are exercised on the same dataset, the same window size ($N = 128$), both the stratified window-level 3-fold protocol and a grouped subject-out 3-fold protocol ($N = 8\,172$ pooled test windows in each), the same classifier panel, and the same set of pointwise, polynomial-trend, and classification metrics. A symmetric evaluation across all four encodings – under two complete 3-fold cross-validations, one window-level and one subject-out – is what allows the comparison reported here to be read as a direct ranking rather than as a partial inventory.

In summary, the four encodings occupy a clear hierarchy under the controlled protocol:

- **GASF** — near-original on every lens and empirically near-equivalent to the raw signal for classification; the established diagonal-inverse baseline and, with GADF, one of the two near-original decoders.
- **GADF** — near-original on every lens; statistically indistinguishable from GASF in classification and marginally the tighter of the two on pointwise error, with a fully specified closed-form decoder.
- **RP** — near-lossless once the inverse normalisation is corrected (MAE ≈ 0.015, Pearson ≈ 1.0); its reconstructions match the original-signal classification accuracy across all six classifiers, placing it with GASF/GADF as a near-original decoder, and it remains the natural choice when recurrence geometry itself is the object of study.
- **MTF** — the only lossy decoder point-to-point; its value is the compact quantile-transition structure it preserves for classification rather than faithful amplitude reconstruction.
- **Scope** — all decoders assume the image is the deterministic output of a known encoder with known settings; RP additionally requires the stored per-axis RP-matrix range for its normalised reconstruction, while the per-axis amplitude range is needed by any method only for physical-unit recovery. This is controlled reconstruction, not a demonstration of generative augmentation.

Within the tested protocol, GASF and GADF are two of the three near-original pointwise decoders: GADF attains a lower observed global MAE than GASF (0.0157 vs 0.0205 over all 8172 windows), while GASF is the established diagonal-inverse baseline (Pearson 0.99988 ± 0.00014), and both match the originals within 0.001 absolute classification accuracy across the six classifiers on the 3-fold mean. The paired McNemar tests detect no GASF or GADF degradation in the evaluated classifiers, and a subject-clustered paired TOST confirms equivalence to the original within a ± 0.02 (two-point) accuracy margin on all six classifiers (Section 7.5); this is an equivalence of downstream classification accuracy for this task rather than a proof of formal information preservation. The restricted-branch, multi-pivot, boundary-anchored GADF decoder is statistically indistinguishable from GASF *in classification* (no detected difference from the originals or from GASF, $p_{\text{Holm}} = 1.0$ throughout, and equivalent within the ± 0.02 margin at the subject

level) and effectively ties it on the polynomial-trend metric (cosine similarity 1.00000 vs 0.99975), while on pointwise error it is marginally the tighter of the two (Pearson 0.99999 ± 0.00001, global MAE 0.0157); together these confirm that the identically zero GADF diagonal does not preclude reconstruction from its off-diagonal entries once an angular anchor, a sign convention, and a restricted branch are fixed (scale metadata being required only to return to physical units). The two Gramian variants are near-identical rather than one dominating the other: the paired per-window Wilcoxon on MAE ($N = 8\,172$ windows) and the subject-clustered Wilcoxon that treats the $N = 51$ subjects as the independent units (per-subject mean MAE 0.0148 for GADF vs 0.0199 for GASF) both reject equality in GADF's favour (Section 7.3); the per-window standardised effect is medium ($r = 0.457$) while the absolute MAE difference remains very small, so neither Gramian variant is meaningfully more faithful than the other in this protocol.

With the corrected decoder, RP is no longer amplitude-lossy: it reconstructs near-losslessly (MAE $= 0.015 \pm 0.02$, Pearson ≈ 1.0; Table 18) and its reconstructions drive all six classifiers to essentially the original-signal accuracy ($\Delta\text{acc} \approx 0$; Table 9), placing it alongside the two Gramian fields. This leaves MTF, under the Fiedler-vector decoder of Section 6.7, as the only amplitude-lossy decoder (MAE $= 3.94 \pm 4.25$, Pearson 0.014 ± 0.49): its Q=4 quantisation is information-theoretically lower-bounded by the within-bin variance, so it retains only the coarse quantile-transition structure of the Markov representation. The ranking is therefore no longer lens-dependent for RP — it is faithful on both the pointwise and the classifier lens — and MTF is the single encoding whose value is as a compact, classifier-oriented representation rather than a faithful inverse.

These results invite three follow-ups. First, the robustness of each decoder under perturbations of the encoded image – additive noise, JPEG/PNG re-quantisation, partial occlusion – is the natural test of the augmentation hypothesis and is the missing piece for a generative-augmentation claim. Second, the GADF angular anchor is identified by a deterministic extreme-mean rule that is exact in the noise-free restricted regime and misidentified no window on WISDM; a principled or learned selection rule, together with sweeps over α, anchor-error rates, and image-noise levels, would characterise its robustness under perturbed and generated images without leaving the closed-form regime. True-vs-global-vs-predicted min–max calibration would quantify the robustness of *optional* physical-scale recovery without altering the normalised inverse, which the decoder already recovers from the image alone. Third, the one remaining lossy decoder, MTF, loses information at a rate lower-bounded by its within-bin variance; whether a single stored sign bit (to remove the global mirror ambiguity) or a finer, possibly adaptive, quantisation can turn it into a near-lossless inverse without sacrificing the compactness that makes it a useful classifier-oriented representation is a natural follow-up. We expect that resolving these three follow-ups in turn would clarify when reconstruction-faithful encodings can substitute for the raw signal in downstream pipelines and when they cannot.

A. Per-Axis Reconstruction Errors

Table 7 in the main text reports the per-window global reconstruction error (averaged over the three accelerometer axes) for each encoder. This appendix gives the per-axis breakdown over the same $N = 8172$ pooled windows for RP (Table 18), MTF (Table 19), GASF (Table 20) and GADF (Table 21); in each, the Global column reproduces the per-window global figure of Table 7.

Table 18: RP per-axis reconstruction errors over all $N = 8172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**Dim 0**	**Dim 1**	**Dim 2**	**Global**
Mean Absolute Error	0.0148 ± 0.0214	0.0183 ± 0.0264	0.0114 ± 0.0153	0.0148 ± 0.0170
Mean Relative Error	0.0037 ± 0.0063	0.0052 ± 0.0066	0.0053 ± 0.0065	0.0047 ± 0.0050
Mean Squared Error	0.0010 ± 0.0076	0.0015 ± 0.0090	0.0005 ± 0.0027	0.0010 ± 0.0046
Pearson Correlation Coefficient	0.99998 ± 0.00009	0.99998 ± 0.00005	0.99998 ± 0.00006	0.99998 ± 0.00005

Table 19: MTF per-axis reconstruction errors over all $N = 8172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**Dim 0**	**Dim 1**	**Dim 2**	**Global**
Mean Absolute Error	4.13 ± 5.11	4.18 ± 5.32	3.53 ± 4.66	3.94 ± 4.25
Mean Relative Error	0.90 ± 1.28	1.28 ± 1.63	1.47 ± 1.51	1.22 ± 1.20
Mean Squared Error	57.39 ± 123.57	61.72 ± 157.60	47.18 ± 158.11	55.43 ± 113.46
Pearson Correlation Coefficient	-0.00214 ± 0.84525	0.03600 ± 0.83941	0.00680 ± 0.82844	0.01356 ± 0.48870

Table 20: GASF per-axis reconstruction errors over all $N = 8172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**Dim 0**	**Dim 1**	**Dim 2**	**Global**
Mean Absolute Error	0.0236 ± 0.0243	0.0195 ± 0.0223	0.0183 ± 0.0216	0.0205 ± 0.0199
Mean Relative Error	0.0056 ± 0.0068	0.0045 ± 0.0045	0.0082 ± 0.0098	0.0061 ± 0.0056
Mean Squared Error	0.0050 ± 0.0107	0.0048 ± 0.0127	0.0030 ± 0.0121	0.0043 ± 0.0088
Pearson Correlation Coefficient	0.99987 ± 0.00020	0.99990 ± 0.00024	0.99987 ± 0.00027	0.99988 ± 0.00014

Table 21: GADF per-axis reconstruction errors over all $N = 8172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**Dim 0**	**Dim 1**	**Dim 2**	**Global**
Mean Absolute Error	0.0159 ± 0.0189	0.0169 ± 0.0215	0.0144 ± 0.0199	0.0157 ± 0.0167
Mean Relative Error	0.0039 ± 0.0057	0.0055 ± 0.0074	0.0070 ± 0.0088	0.0055 ± 0.0057
Mean Squared Error	0.0008 ± 0.0020	0.0009 ± 0.0028	0.0007 ± 0.0028	0.0008 ± 0.0018
Pearson Correlation Coefficient	0.99999 ± 0.00002	0.99999 ± 0.00002	0.99999 ± 0.00002	0.99999 ± 0.00001

B. Per-Class Classification Results

This appendix reports the complete per-class precision, recall, and F1-score results for the six evaluated classifiers. The main text retains only the compact accuracy comparison required to interpret the differences among reconstruction methods. Per-class Support values are per-class fold-means rounded to integers and may sum to 2723 rather than the 2724 windows per fold reported in the Accuracy row (independent rounding).

Table 22: Classification Results for GRU on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.89±0.03	0.93±0.01	0.91±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544	0.65±0.01	0.61±0.04	0.63±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544
1	0.99±0.01	0.99±0.00	0.99±0.00	532	0.99±0.01	0.99±0.00	0.99±0.00	532	0.75±0.02	0.99±0.00	0.85±0.01	532	0.99±0.01	0.99±0.00	0.99±0.00	532	0.99±0.01	0.99±0.00	0.99±0.00	532
2	0.91±0.01	0.86±0.03	0.88±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536	0.57±0.01	0.52±0.03	0.54±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536
3	0.88±0.01	0.96±0.01	0.92±0.00	551	0.88±0.01	0.96±0.01	0.92±0.00	551	0.90±0.01	0.90±0.01	0.90±0.01	551	0.88±0.01	0.96±0.01	0.92±0.00	551	0.88±0.01	0.96±0.01	0.92±0.00	551
4	0.95±0.01	0.89±0.01	0.92±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560	0.96±0.01	0.81±0.01	0.88±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560
Accuracy	0.92±0.00 (2724)				0.92±0.00 (2724)				0.77±0.01 (2724)				0.92±0.00 (2724)				0.92±0.00 (2724)			
Macro Avg	0.93±0.00/0.92±0.00/0.92±0.00				0.92±0.00/0.92±0.00/0.92±0.00				0.77±0.01/0.76±0.01/0.76±0.01				0.92±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00			
Weighted Avg	0.93±0.00/0.92±0.00/0.92±0.00				0.92±0.00/0.92±0.00/0.92±0.00				0.77±0.01/0.77±0.01/0.76±0.01				0.92±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00			

Table 23: Classification Results for KNN on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.74±0.01	0.52±0.01	0.61±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544	0.64±0.01	0.49±0.02	0.55±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544
1	1.00±0.00	0.75±0.01	0.86±0.01	532	1.00±0.00	0.75±0.01	0.86±0.01	532	0.99±0.02	0.37±0.00	0.54±0.00	532	1.00±0.00	0.75±0.01	0.86±0.01	532	1.00±0.00	0.75±0.01	0.86±0.01	532
2	0.49±0.04	0.04±0.01	0.08±0.02	536	0.49±0.04	0.04±0.01	0.08±0.01	536	0.43±0.04	0.07±0.00	0.12±0.00	536	0.50±0.04	0.04±0.01	0.08±0.02	536	0.50±0.04	0.04±0.01	0.08±0.01	536
3	0.79±0.00	0.96±0.00	0.87±0.00	551	0.79±0.00	0.96±0.00	0.87±0.00	551	0.63±0.01	0.94±0.01	0.75±0.01	551	0.79±0.00	0.96±0.00	0.87±0.00	551	0.79±0.00	0.96±0.00	0.87±0.00	551
4	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.01	0.90±0.01	0.57±0.01	560	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.00	0.92±0.01	0.58±0.00	560
Accuracy	0.64±0.00 (2724)				0.64±0.00 (2724)				0.56±0.01 (2724)				0.64±0.00 (2724)				0.64±0.00 (2724)			
Macro Avg	0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00				0.62±0.01/0.55±0.01/0.51±0.01				0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00			
Weighted Avg	0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00				0.62±0.01/0.56±0.01/0.51±0.01				0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00			

Table 24: Classification Results for CNN on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.89±0.05	0.89±0.05	0.89±0.01	544	0.91±0.02	0.85±0.05	0.87±0.02	544	0.61±0.07	0.73±0.08	0.66±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544
1	0.99±0.01	0.98±0.00	0.99±0.00	532	0.98±0.01	0.99±0.00	0.99±0.00	532	0.88±0.03	0.97±0.02	0.92±0.01	532	0.99±0.01	0.98±0.00	0.99±0.00	532	0.99±0.01	0.98±0.00	0.99±0.00	532
2	0.88±0.04	0.87±0.05	0.87±0.01	536	0.84±0.04	0.90±0.03	0.86±0.01	536	0.62±0.03	0.55±0.10	0.57±0.05	536	0.88±0.04	0.87±0.05	0.87±0.01	536	0.88±0.04	0.87±0.05	0.87±0.01	536
3	0.84±0.05	0.97±0.02	0.90±0.02	551	0.84±0.03	0.97±0.01	0.90±0.01	551	0.85±0.04	0.89±0.04	0.87±0.01	551	0.84±0.04	0.97±0.02	0.90±0.02	551	0.84±0.05	0.97±0.02	0.90±0.02	551
4	0.97±0.02	0.83±0.05	0.89±0.02	560	0.98±0.00	0.81±0.02	0.88±0.01	560	0.97±0.03	0.73±0.04	0.83±0.02	560	0.97±0.02	0.83±0.05	0.89±0.02	560	0.97±0.02	0.82±0.05	0.89±0.02	560
Accuracy	0.91±0.00 (2724)				0.90±0.01 (2724)				0.77±0.01 (2724)				0.91±0.01 (2724)				0.91±0.00 (2724)			
Macro Avg	0.91±0.00/0.91±0.00/0.91±0.00				0.91±0.00/0.90±0.01/0.90±0.01				0.79±0.01/0.77±0.01/0.77±0.01				0.91±0.00/0.91±0.01/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00			
Weighted Avg	0.91±0.00/0.91±0.00/0.91±0.01				0.91±0.00/0.90±0.01/0.90±0.01				0.79±0.01/0.77±0.01/0.77±0.01				0.91±0.00/0.91±0.01/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00			

Table 25: Classification Results for Random Forest on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.85±0.01	0.79±0.00	0.82±0.00	544	0.86±0.01	0.79±0.00	0.82±0.00	544	0.68±0.01	0.60±0.01	0.64±0.00	544	0.85±0.01	0.79±0.01	0.82±0.01	544	0.85±0.01	0.79±0.00	0.82±0.00	544
1	0.94±0.00	0.99±0.00	0.96±0.00	532	0.94±0.00	0.99±0.00	0.96±0.00	532	0.65±0.01	1.00±0.00	0.79±0.01	532	0.94±0.00	0.99±0.00	0.96±0.00	532	0.94±0.00	0.99±0.00	0.96±0.00	532
2	0.79±0.00	0.82±0.01	0.81±0.01	536	0.79±0.00	0.83±0.01	0.81±0.01	536	0.63±0.01	0.51±0.02	0.56±0.01	536	0.79±0.01	0.82±0.01	0.80±0.01	536	0.79±0.00	0.82±0.01	0.80±0.01	536
3	0.92±0.01	0.95±0.00	0.93±0.01	551	0.92±0.01	0.95±0.00	0.93±0.01	551	0.93±0.01	0.89±0.01	0.91±0.00	551	0.92±0.01	0.95±0.00	0.93±0.01	551	0.92±0.01	0.95±0.00	0.93±0.00	551
4	0.96±0.00	0.90±0.01	0.93±0.00	560	0.96±0.00	0.90±0.01	0.93±0.00	560	0.94±0.01	0.79±0.01	0.86±0.01	560	0.96±0.00	0.90±0.01	0.93±0.00	560	0.96±0.00	0.90±0.01	0.93±0.00	560
Accuracy	0.89±0.00 (2724)				0.89±0.00 (2724)				0.76±0.00 (2724)				0.89±0.00 (2724)				0.89±0.00 (2724)			
Macro Avg	0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.77±0.00/0.76±0.00/0.75±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00			
Weighted Avg	0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.77±0.00/0.76±0.00/0.75±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00			

Table 26: Classification Results for Support Vector Machine on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.76±0.01	0.65±0.02	0.70±0.01	544	0.76±0.01	0.65±0.01	0.70±0.01	544	0.52±0.01	0.71±0.02	0.60±0.01	544	0.76±0.01	0.65±0.01	0.70±0.01	544	0.76±0.00	0.65±0.01	0.70±0.01	544
1	0.97±0.01	0.97±0.00	0.97±0.00	532	0.97±0.01	0.97±0.00	0.97±0.00	532	0.69±0.01	1.00±0.00	0.81±0.01	532	0.97±0.01	0.97±0.00	0.97±0.00	532	0.97±0.01	0.97±0.00	0.97±0.00	532
2	0.66±0.01	0.69±0.01	0.67±0.01	536	0.66±0.01	0.69±0.01	0.67±0.01	536	0.49±0.04	0.17±0.01	0.25±0.02	536	0.65±0.01	0.69±0.01	0.67±0.01	536	0.66±0.01	0.69±0.01	0.67±0.01	536
3	0.84±0.01	0.84±0.00	0.84±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551	0.86±0.01	0.81±0.01	0.83±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551
4	0.78±0.01	0.85±0.01	0.81±0.00	560	0.78±0.01	0.85±0.01	0.81±0.01	560	0.86±0.00	0.81±0.00	0.83±0.00	560	0.78±0.01	0.85±0.01	0.81±0.00	560	0.78±0.00	0.85±0.01	0.81±0.00	560
Accuracy	0.80±0.00 (2724)				0.80±0.00 (2724)				0.70±0.01 (2724)				0.80±0.00 (2724)				0.80±0.00 (2724)			
Macro Avg	0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.01/0.80±0.00/0.80±0.00				0.69±0.01/0.70±0.01/0.67±0.01				0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00			
Weighted Avg	0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.01/0.80±0.00/0.80±0.00				0.69±0.01/0.70±0.01/0.67±0.01				0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00			

Table 27: Classification Results for LSTM on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.80±0.11	0.85±0.06	0.81±0.04	544	0.79±0.11	0.85±0.06	0.81±0.04	544	0.62±0.06	0.65±0.10	0.63±0.02	544	0.79±0.10	0.85±0.06	0.81±0.04	544	0.80±0.11	0.85±0.07	0.81±0.04	544
1	0.98±0.02	0.98±0.01	0.98±0.01	532	0.98±0.02	0.98±0.01	0.98±0.01	532	0.77±0.01	0.96±0.02	0.85±0.02	532	0.98±0.02	0.98±0.01	0.98±0.01	532	0.98±0.02	0.98±0.01	0.98±0.01	532
2	0.83±0.05	0.71±0.21	0.74±0.12	536	0.83±0.05	0.71±0.21	0.74±0.12	536	0.61±0.03	0.46±0.18	0.50±0.13	536	0.83±0.04	0.71±0.20	0.74±0.12	536	0.83±0.05	0.71±0.21	0.74±0.12	536
3	0.87±0.01	0.94±0.01	0.90±0.00	551	0.87±0.01	0.94±0.01	0.90±0.00	551	0.87±0.00	0.92±0.00	0.89±0.00	551	0.87±0.00	0.94±0.01	0.90±0.00	551	0.87±0.01	0.94±0.01	0.90±0.00	551
4	0.92±0.04	0.86±0.02	0.89±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560	0.95±0.01	0.79±0.02	0.86±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560
Accuracy	0.87±0.03 (2724)				0.87±0.03 (2724)				0.76±0.02 (2724)				0.87±0.03 (2724)				0.87±0.03 (2724)			
Macro Avg	0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04				0.76±0.01/0.76±0.02/0.75±0.03				0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04			
Weighted Avg	0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04				0.76±0.01/0.76±0.02/0.75±0.03				0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04			

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